
Internship Report

Subject:
Power series:
from the boundary of the disc to the Catalan numbers

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Synopsis

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Title of the internship: From power series to generating functions: convergence on the boundary and applications

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This internship is devoted to the study of power series and their various applications. First, we take a closer look at a question raised in lectures: if a power series converges on the unit circle, does it converge uniformly on the closed disc?

Next, we explore the use of power series in probability theory, in particular through the study of Galton–Watson processes and random walks.

Finally, we turn to their applications in economics, especially in discounting models and risk theory.

This internship also gave me the opportunity to write a document in $\text{\LaTeX}$, which is very useful for producing reports, CVs, cover letters, and so on.

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Chapter 1

Behaviour of a power series on its circle of convergence

1.1 Introduction

In the study of power series, the behaviour inside the open disc of convergence is governed by a strong form of regularity. Indeed, we know that convergence there is absolute, and normal on every disc contained in the first one. However, the study on the exact boundary of that domain (the circle) brings out far more subtle analytic phenomena.

A classical question raises the fundamental distinction between pointwise and uniform convergence at the edge of the domain. The usual theorem guarantees that the limit of a sequence of continuous functions (such as the polynomials formed by the partial sums of a power series) which converges uniformly is itself continuous. From there, an interesting problem arises: is it possible to construct a power series that converges pointwise at every point of the unit circle, but whose convergence is not uniform on the closed disc?

This section sets out to study that phenomenon by taking explicit counterexamples apart; we will then bring to light how pointwise convergence can be preserved at every single point while still generating an unbounded limit function.

1.2 Reminders

First of all, I will recall various definitions concerning power series and prove, whenever this was not already done in the lectures, the various tools that I will need later on.

Definition 1.1. A **power series** is any series of the form

$$\sum_{n=0}^{\infty} a_n z^n,$$

where $(a_n)_{n \geq 0}$ is a sequence of complex numbers.

Definition 1.2. The limit superior of the sequence $(a_n)_{n \geq 0}$, a sequence of real numbers for every $n \geq 0$, is the number L written $\limsup(u_n)$ and defined by

$$L = \lim_{n \rightarrow \infty} (\sup_{k \geq n} u_k) \text{ with } L \in \mathbb{R} \cup \{\pm\infty\}$$

Property 1.1. If $\lim(u_n) \in \mathbb{R} \cup \{\pm\infty\}$, then $\lim(u_n) = \limsup(u_n)$.

Theorem 1.1. Let $(u_n)_{n \geq 0}$ be a sequence in $\mathbb{R}$:

if $\limsup(u_n) < 1$ then $\sum_{n \geq 0} u_n$ converges;

if $\limsup(u_n) > 1$ then $\sum_{n \geq 0} u_n$ diverges coarsely.

Theorem 1.2. Let $\sum_{n \geq 0} a_n z^n$ be a power series. There exists a unique number, called the radius of convergence and written $R \in [0, +\infty]$ with $R = 1/L$, such that:

1. For every z with $|z| < R$, the series converges absolutely.

2. For every z with $|z| > R$, the series diverges coarsely.

Lemma 1.1. Let $\sum_{n \geq 0} a_n z^n$ be a power series with radius of convergence $R > 0$. For every $r < R$, the series $\sum_{n \geq 0} a_n z^n$ converges uniformly on the closed disc $\overline{D(0, r)}$. In particular, there exist $C > 0$ and $\rho < 1$ such that

$$\sup_{z \in \overline{D(0, r)}} |a_n z^n| \leq C \rho^n$$

Theorem 1.3 (Term-by-term differentiation, reminder). Let $\sum_{n \geq 0} a_n t^n$ be a power series with radius of convergence $R > 0$ and sum $S(t)$ on $] - R, R[$. Then S is of class $\mathcal{C}^\infty$ on $] - R, R[$ and, for every $p \in \mathbb{N}$ and every $t \in] - R, R[$,

$$S^{(p)}(t) = \sum_{n \geq p} n(n-1) \cdots (n-p+1) a_n t^{n-p},$$

this differentiated series also having radius of convergence R .

Proposition 1.1. Let X be a non-empty subset of $\mathbb{C}$, and let $(g_k)_{k \geq 0}$ be a sequence of bounded functions from X to $\mathbb{C}$ converging uniformly on X to a function g . Then g is bounded on X .

Proof. By the definition of uniform convergence, there exists K_0 such that $\sup_{x \in X} |g_{K_0}(x) - g(x)| \leq 1$. Since the sequence of functions is bounded on X , there exists $M > 0$ such that $|g_{K_0}| \leq M$. Then for every $x \in X$:

$$|g(x)| = |g(x) - g_{K_0}(x) + g_{K_0}(x)| \underset{\text{triangle ineq.}}{\leq} |g(x) - g_{K_0}(x)| + |g_{K_0}(x)| \leq 1 + M,$$

so g is bounded on X . □

Theorem 1.4 (Abel, radial limit). Let $\sum_{n \geq 0} a_n z^n$ be a power series with radius of convergence $R \geq 1$. Assume that the series *converges* at the point $z = 1$, that is, that

$\sum_{n \geq 0} a_n$ converges, with sum $S = \sum_{n=0}^{\infty} a_n$. Then the function $f(x) = \sum_{n=0}^{\infty} a_n x^n$, defined on $[0, 1]$, is continuous at 1 from the left:

$$\lim_{x \rightarrow 1^-} \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} a_n = S.$$

Proof. Step 0: reduction. Replacing a_0 by $a_0 - S$ if necessary, we may assume $S = 0$, that is, $\sum_{n=0}^{\infty} a_n = 0$. Indeed, this translates f by the constant S and the sum by S , without changing the nature of the problem. We then have to show that $f(x) \rightarrow 0$ as $x \rightarrow 1^-$.

Let us introduce the *remainders* of the series at 1:

$$R_n = \sum_{k=n+1}^{\infty} a_k, \quad \text{so that} \quad R_{-1} = \sum_{k=0}^{\infty} a_k = 0.$$

Since $\sum a_k$ converges to 0, we have $R_n \rightarrow 0$ as $n \rightarrow \infty$. We also write $A_n = \sum_{k=0}^n a_k$ for the partial sums; then $A_n = -R_n$ (because $A_n + R_n = 0$), so $A_n \rightarrow 0$ as well.

Step 1: rewriting $f(x)$ by Abel summation. Fix $x \in [0, 1[$. The series $\sum a_n x^n$ converges absolutely (because $|x| < 1 \leq R$). Apply Abel's summation by parts with $u_n = a_n$ and $v_n = x^n$, starting at $p = 0$ and letting $q \rightarrow \infty$. The partial sums of u_n are $U_n = A_n$, and the boundary term $U_q v_q = A_q x^q \rightarrow 0$ as $q \rightarrow \infty$ (because $A_q \rightarrow 0$ and $|x^q| \leq 1$). This gives

$$f(x) = \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} A_n (x^n - x^{n+1}) = (1-x) \sum_{n=0}^{\infty} A_n x^n.$$

We can rewrite this with the remainders, using $A_n = -R_n$:

$$f(x) = -(1-x) \sum_{n=0}^{\infty} R_n x^n.$$

(The series $\sum A_n x^n$ does converge for $|x| < 1$, since (A_n) is bounded, being convergent.)

Step 2: estimation. Let $\varepsilon > 0$. Since $R_n \rightarrow 0$, there exists N such that $|R_n| \leq \varepsilon$ for every $n \geq N$. Let us split the sum into two parts:

$$|f(x)| = (1-x) \left| \sum_{n=0}^{\infty} R_n x^n \right| \leq (1-x) \sum_{n=0}^{N-1} |R_n| x^n + (1-x) \sum_{n=N}^{\infty} |R_n| x^n.$$

The tail ($n \geq N$) is bounded using $|R_n| \leq \varepsilon$ and the geometric series $\sum_{n \geq 0} x^n = \frac{1}{1-x}$:

$$(1-x) \sum_{n=N}^{\infty} |R_n| x^n \leq (1-x) \varepsilon \sum_{n=0}^{\infty} x^n = (1-x) \varepsilon \frac{1}{1-x} = \varepsilon.$$

The head ($0 \leq n \leq N-1$) is a finite sum. Set $M = \max_{0 \leq n \leq N-1} |R_n|$ (a constant, independent of x). Then, bounding $x^n \leq 1$,

$$(1-x) \sum_{n=0}^{N-1} |R_n| x^n \leq (1-x) N M \xrightarrow{x \rightarrow 1^-} 0.$$

There therefore exists $\delta > 0$ such that this term is $\leq \varepsilon$ as soon as $1 - \delta < x < 1$.

Conclusion. For every $x \in]1 - \delta, 1[$, we have $|f(x)| \leq \varepsilon + \varepsilon = 2\varepsilon$. Since $\varepsilon > 0$ is arbitrary, this shows that $f(x) \rightarrow 0$ as $x \rightarrow 1^-$, that is (returning to the general case) $f(x) \rightarrow S$. □

1.3 An attempt with classical series

As a reminder, our question is: **if a series converges on the unit circle, will it converge uniformly on the closed disc?**

Our first instinct would be to use the series

$$\sum_{n \geq 0} \frac{z^n}{n}.$$

However, at $z = 1$ we obtain $\sum_{n \geq 0} \frac{1}{n}$, which diverges since it is the harmonic series.

We might then think that adding up terms that are all positive will not work. Let $(a_n)_n$ be a numerical sequence with for all $n \in \mathbb{N}$, $a_n \geq 0$; if $\sum_{n \geq 0} a_n$ converges, then we have

$$\begin{aligned} \text{For every } z \text{ with } |z| < R, \quad & \sum_{n \geq 0} |a_n z^n| < \sum_{n \geq 0} a_n \\ \Rightarrow \sum_{n \geq 0} \|a_n z^n\|_\infty & < \sum_{n \geq 0} a_n \\ \Rightarrow \sum_{n \geq 0} a_n z^n & \text{ converges normally on the disc } \overline{D(0, R)}. \end{aligned}$$

We could then try to fix this problem by producing an alternating series. So let us try the series

$$\sum_{n \geq 0} (-1)^n \frac{z^n}{n}.$$

This series might make us believe it is a good example, because at $z = 1$ we obtain $\sum_{n \geq 0} \frac{(-1)^n}{n}$, which is an alternating series and therefore converges at 1 by the alternating series test. However, at $z = -1$ we again obtain $\sum_{n \geq 0} \frac{1}{n}$, which diverges.

We see that this alternating series $(+, -, +, -, \dots)$ does not work. But is there a series with a sign pattern (for instance $+, +, -, +, +, -$) that answers our question?

Let $(a_n)_n$ be a real sequence following such a sign pattern. Suppose that the series converges at the m -th roots of unity $\omega_0, \omega_1, \dots, \omega_{m-1}$ (with $\omega_p = e^{\frac{2ip\pi}{m}}$ for every $p \in \llbracket 0, m-1 \rrbracket$), where $m \geq 1$ is an integer.

Let me define the k -th block evaluated at z :

$$B_k(z) = \sum_{j=0}^{m-1} a_{mk+j} z^{mk+j}$$

Evaluating at each root of unity, and knowing that $(\omega_p)^{mk} = 1$, we obtain

$$\forall p \in \llbracket 0, m-1 \rrbracket, \quad B_k(\omega_p) = \sum_{j=0}^{m-1} a_{mk+j} \omega_p^j$$

Now we know that

$$\sum_{p=0}^{m-1} \omega_p^{j-r} = \begin{cases} m & \text{if } j = r \\ 0 & \text{otherwise} \end{cases} \quad (j, r \in \llbracket 0, m-1 \rrbracket)$$

We may sum the evaluated blocks over p after multiplying them by ω_p^{-r} , so as to isolate the term of inner index r ; this yields

$$\sum_{p=0}^{m-1} \omega_p^{-r} B_k(\omega_p) = m a_{mk+r}$$

By assumption, the series $\sum_n a_n \omega_p^n$ converges at each ω_p ; since a block gathers m consecutive terms, the m block series $\sum_k B_k(\omega_p)$ are convergent as well. As the space of convergent series is a vector space, every linear combination of these series is again a convergent series. Summing the previous identity over k , we therefore obtain, for each inner index r :

$$\sum_{k=0}^{\infty} a_{mk+r} = \frac{1}{m} \sum_{p=0}^{m-1} \omega_p^{-r} \sum_{k=0}^{\infty} B_k(\omega_p) \quad \text{converges.}$$

We are then in a position to build the series of absolute values $|a_n|$: the convergence of $\sum a_{mk+r}$ ensures the absolute convergence of $\sum |a_{mk+r}|$, since this subsequence

keeps a constant sign modulo m . Summing over the m classes:

$$\sum_{n=0}^{\infty} |a_n| = \sum_{r=0}^{m-1} \left(\sum_{k=0}^{\infty} |a_{mk+r}| \right) < +\infty$$

Now, for every point z in the closed unit disc, we can bound the general term of our power series in modulus:

$$\sup_{|z| \leq 1} |a_n z^n| = |a_n|,$$

the general term of a convergent series. We therefore obtain normal convergence, since $\sum |a_n|$ converges. In conclusion, if the sign of a_n depends exclusively on $n \pmod{m}$, then convergence at the m points $e^{2ip\pi/m}$ forces the absolute convergence of the series of the $|a_n|$, hence uniform convergence on the closed unit disc.

1.4 Historical background

Our question already appears at the beginning of the 20th century, in the work of Pringsheim and Fejér on the behaviour of power series at the circle of convergence. In 1910, answering a question of Pringsheim, Fejér constructed a first power series of radius 1 whose partial sums converge at every point of the unit circle, but in such an irregular way that the convergence is not uniform on the closed disc.¹ That construction already relies on a decomposition into blocks and on sharp estimates of trigonometric sums, very close to those used in the theory of Fourier series.

A few years later, Hardy and Littlewood took up this theme in a different context, that of Diophantine approximation and theta functions. They studied series of the type

¹L. Fejér, *Über gewisse Potenzreihen an der Konvergenzgrenze*, Sitzungsberichte der math.-phys. Klasse der K. Bayer. Akad. der Wiss., 1910.

$$\sum_{n \geq 0} n^{-\beta} e^{icn^\alpha} z^n,$$

and showed that, for a certain pair (α, β) , one obtains precisely power series that converge at every point of the unit circle but not uniformly on that circle.² The Hardy–Littlewood example belonging to this family is

$$\sum_{n \geq 0} n^{-3/4} e^{i\sqrt{n}} z^n.$$

More recently, Lowther became interested in the same kind of questions, but from a slightly different angle. His aim is to show very concretely that, even if a power series behaves very well inside its disc of convergence, nothing guarantees equally regular behaviour on the boundary. To do so, he starts from a function defined as a sum of simple fractions

$$f(z) = \sum_{n \geq 1} \frac{1}{n^5(\alpha_n - z)}, \quad \alpha_n = 1 + \frac{i}{n^3},$$

which converges normally on every disc $\overline{D(0, r)}$ with $r < 1$ and therefore defines a holomorphic function on the unit disc. Expanding f as a Taylor series around 0, one obtains a power series with radius of convergence 1 that converges at every point of the unit circle, but whose convergence is not uniform on the closed disc: the poles α_n , accumulating towards $z = 1$, force the limit function to become very large near that point, which rules out any uniform control up to the boundary.

To get hold of these examples, I did not start from scratch: I made extensive use of artificial intelligence assistants (in particular Gemini and Claude). These models are trained on a very large amount of mathematical text available online, which

²G. H. Hardy, J. E. Littlewood, *Some problems of Diophantine approximation II*, Acta Math. 37 (1914), 155–191.

in practice makes them excellent interactive encyclopaedias. They allowed me to identify quickly the papers of Fejér and Hardy–Littlewood, as well as Lowther’s work.

In the next section, I will prove that Lowther’s power series answers our question.

1.5 Proof for Lowther’s series

1.5.1 Introduction and preliminaries

Let $\mathbb{D} = \{z \in \mathbb{C} \mid |z| \leq 1\}$ be the closed unit disc and let $\mathbb{U} = \{z \in \mathbb{C} \mid |z| = 1\}$ be the unit circle.

For every $n \geq 1$, set

$$f(z) = \sum_{n \geq 1} \frac{1}{n^5(\alpha_n - z)}, \quad \text{with } \alpha_n = 1 + \frac{i}{n^3},$$

for those $z \in \mathbb{C}$ at which this series converges.

For every $n \geq 1$, $|\alpha_n|^2 = 1^2 + (n^{-3})^2 = 1 + n^{-6} > 1$. For every $z \in \mathbb{D}$, $z \neq \alpha_n$. We also have

$$\lim_{n \rightarrow \infty} |\alpha_n| = 1,$$

hence $\alpha_n \rightarrow 1$.

I will show that the function f is well defined. Fix $z \in \mathbb{D}$.

1. if $z = 1$, then $|\alpha_n - 1| = n^{-3}$, and

$$\frac{1}{|n^5(\alpha_n - 1)|} = \frac{1}{n^5|\alpha_n - 1|} = \frac{1}{n^5 n^{-3}} = \frac{1}{n^2}$$

which is a convergent Riemann series ($\alpha = 2$), so the series converges absolutely and equals

$$f(1) = \sum_{n=1}^{\infty} \frac{1}{n^5(\alpha_n - 1)} = \sum_{n=1}^{\infty} \frac{1}{n^5 i n^{-3}} = \sum_{n=1}^{\infty} \frac{-i}{n^2} = -i \frac{\pi^2}{6} (I)$$

2. if $z \neq 1$, then since $\alpha_n \rightarrow 1$ we have $|\alpha_n - z| \rightarrow |1 - z|$. By the definition of a limit, there exists N such that for $n \geq N$, $|\alpha_n - z| \geq \frac{|1-z|}{2}$. Then

$$\frac{1}{n^5 |\alpha_n - z|} \leq \frac{2}{n^5 |1 - z|},$$

so the series converges absolutely, being dominated by a convergent Riemann series ($\alpha = 5$).

Let $\alpha \in \mathbb{C}$ and let $z \in \mathbb{D}$ with $\alpha \neq z$ and $\alpha \neq 0$. For every integer $K \geq 0$, we have

$$\sum_{k=0}^K \frac{z^k}{\alpha^{k+1}} = \frac{1}{\alpha} \sum_{k=0}^K \frac{z^k}{\alpha^k} = \frac{1}{\alpha} \sum_{k=0}^K \left(\frac{z}{\alpha}\right)^k \underset{\text{since } \frac{z}{\alpha} \neq 1}{=} \frac{1}{\alpha} \frac{1 - \left(\frac{z}{\alpha}\right)^{K+1}}{1 - \frac{z}{\alpha}} = \frac{1}{\alpha - z} \left(1 - \left(\frac{z}{\alpha}\right)^{K+1}\right).$$

Set, for every integer $k \geq 0$: $a_k := \sum_{n=1}^{\infty} \frac{1}{n^5 \alpha_n^{k+1}}$

$$\begin{aligned} \frac{1}{n^5 |\alpha_{k+1}|} &\leq \frac{1}{n^5} \\ \Rightarrow |a_k| &\leq \sum_{n \geq 1} \frac{1}{n^5} \leq \sum_{n \geq 1} \frac{1}{n^2} \leq 2. \end{aligned}$$

The series a_k converges absolutely and, by the lectures, the power series $\sum_{k \geq 1} a_k z^k$ has a radius of convergence R satisfying $R \geq 1$.

Let $z \in \mathbb{D}$ and $K \geq 0$. Write $S_K(z) = \sum_{k=0}^K a_k z^k$ for the partial sum of the power series. Using the definition of the a_k :

$$S_K(z) = \sum_{k=0}^K z^k \sum_{n=1}^{\infty} \frac{1}{n^5 \alpha_n^{k+1}} = \sum_{n=1}^{\infty} \frac{1}{n^5} \sum_{k=0}^K \frac{z^k}{\alpha_n^{k+1}}.$$

Exchanging the two sums is allowed because the series a_k converges and the sum over k is finite.

Applying the geometric formula established above with $\alpha = \alpha_n$:

$$S_K(z) = \sum_{n=1}^{\infty} \frac{1}{n^5(\alpha_n - z)} \left(1 - \left(\frac{z}{\alpha_n} \right)^{K+1} \right) = f(z) - R_K(z),$$

with

$$R_K(z) = \sum_{n=1}^{\infty} \frac{1}{n^5(\alpha_n - z)} \left(\frac{z}{\alpha_n} \right)^{K+1}.$$

Splitting into two series is justified because each of them converges absolutely: that of $f(z)$ as shown above, and that of $R_K(z)$ by the bound

$$\left| \frac{1}{n^5(\alpha_n - z)} \left(\frac{z}{\alpha_n} \right)^{K+1} \right| \leq \frac{1}{n^5 |\alpha_n - z|},$$

since $|z| \leq 1 < |\alpha_n|$ implies $|z/\alpha_n| \leq 1$.

1.5.2 Convergence at every point of the closed disc

Fix $z \in \mathbb{D}$. Let us show that $R_K(z) \rightarrow 0$ as $K \rightarrow \infty$. Set $b_n = \frac{1}{n^5 |\alpha_n - z|}$; by the study of the existence of f , the series $\sum_n b_n$ converges. Let $\varepsilon > 0$.

Since $\sum_n b_n$ converges, there exists an index n_0 such that $\sum_{n>n_0} b_n \leq \frac{\varepsilon}{2}$. Since $|z/\alpha_n| \leq 1$, we have for every K :

$$\left| \sum_{n>n_0} \frac{1}{n^5(\alpha_n - z)} \left(\frac{z}{\alpha_n} \right)^{K+1} \right| \leq \sum_{n>n_0} b_n \leq \frac{\varepsilon}{2}.$$

For the remaining part, set $q = \max_{1 \leq n \leq n_0} \frac{1}{|\alpha_n|}$: the maximum of finitely many reals that are all strictly less than 1, so $q < 1$. Then

$$\left| \sum_{n \leq n_0} \frac{1}{n^5(\alpha_n - z)} \left(\frac{z}{\alpha_n} \right)^{K+1} \right| \leq q^{K+1} \sum_{n \leq n_0} b_n \leq q^{K+1} \sum_{n \geq 1} b_n \xrightarrow{K \rightarrow \infty} 0,$$

because $q < 1$. There therefore exists K_0 such that this quantity is $\leq \frac{\varepsilon}{2}$ for $K \geq K_0$. Putting the pieces back together, $|R_K(z)| \leq \varepsilon$ for $K \geq K_0$, hence $R_K(z) \rightarrow 0$.

From the formula $S_K(z) = f(z) - R_K(z)$, we conclude that $S_K(z) \rightarrow f(z)$: the power series $\sum_k a_k z^k$ converges at every point $z \in \mathbb{D}$, with sum $f(z)$.

1.5.3 The function f is not bounded on the circle

Let z_m be the sequence of points of the unit circle defined by

$$z_m = \sqrt{1 - m^{-6}} + \frac{i}{m^3}, \quad m \geq 1.$$

We have $|z_m|^2 = (1 - m^{-6}) + m^{-6} = 1$, so $z_m \in \mathbb{U}$, and $z_m \rightarrow 1$.

Since α_m and z_m have the same imaginary part m^{-3} , their difference is real:

$$\alpha_m - z_m = 1 - \sqrt{1 - m^{-6}} > 0.$$

Multiplying by the conjugate quantity, for $x \in [0, 1]$:

$$1 - \sqrt{1-x} = \frac{1 - (1-x)}{1 + \sqrt{1-x}} = \frac{x}{1 + \sqrt{1-x}},$$

and since $1 + \sqrt{1-x} \in [1, 2]$, we obtain $\frac{x}{2} \leq 1 - \sqrt{1-x} \leq x$. With $x = m^{-6}$:

$$\frac{m^{-6}}{2} \leq \alpha_m - z_m \leq m^{-6}.$$

The term of index $n = m$ in the series defining $f(z_m)$, written $T_m = \frac{1}{m^5(\alpha_m - z_m)}$, therefore satisfies

$$m = \frac{m^6}{m^5} \leq T_m \leq \frac{2m^6}{m^5} = 2m. (II)$$

It remains to control the other terms. For $n \neq m$, we bound our modulus from below so as to obtain

$$|\alpha_n - z_m| \geq |\operatorname{Im}(\alpha_n - z_m)| = \left| \frac{1}{n^3} - \frac{1}{m^3} \right|,$$

and we shall bound $\Sigma_m = \sum_{n \neq m} \frac{1}{n^5 |n^{-3} - m^{-3}|}$ from above by a constant independent of m , splitting into three zones.

(Zone 1): $2n \leq m$. Here $n \leq \frac{m}{2}$, so $\frac{n}{m} \leq \frac{1}{2}$, whence $\left(\frac{n}{m}\right)^3 \leq \frac{1}{8}$. Since $n < m$, we have $n^{-3} > m^{-3}$, and therefore

$$\left| \frac{1}{n^3} - \frac{1}{m^3} \right| = \frac{1}{n^3} - \frac{1}{m^3} = \frac{1}{n^3} \left(1 - \frac{n^3}{m^3} \right) \geq \frac{1}{n^3} \left(1 - \frac{1}{8} \right) = \frac{7}{8} \cdot \frac{1}{n^3}.$$

Carrying this lower bound into the denominator, each term is bounded above by

$$\frac{1}{n^5 |n^{-3} - m^{-3}|} \leq \frac{1}{n^5 \cdot \frac{7}{8} n^{-3}} = \frac{8}{7} \cdot \frac{1}{n^2}.$$

Zone 1 is therefore bounded above by

$$\sum_{2n \leq m} \frac{8}{7} \cdot \frac{1}{n^2} \leq \frac{8}{7} \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{8}{7} \cdot \frac{\pi^2}{6} \leq \frac{8}{7} \cdot 1.645 \leq 2.$$

(Zone 3): $n \geq 2m$. Here $m \leq \frac{n}{2}$, so $\frac{m}{n} \leq \frac{1}{2}$ and $\left(\frac{m}{n}\right)^3 \leq \frac{1}{8}$. Since $n > m$, this time $m^{-3} > n^{-3}$, and symmetrically

$$\left| \frac{1}{n^3} - \frac{1}{m^3} \right| = \frac{1}{m^3} - \frac{1}{n^3} = \frac{1}{m^3} \left(1 - \frac{m^3}{n^3} \right) \geq \frac{7}{8} \cdot \frac{1}{m^3}.$$

Each term is therefore bounded above by

$$\frac{1}{n^5 |n^{-3} - m^{-3}|} \leq \frac{1}{n^5 \cdot \frac{7}{8} m^{-3}} = \frac{8}{7} \cdot \frac{m^3}{n^5}.$$

To sum over $n \geq 2m$, we compare with an integral: the function $f(t) = t^{-5}$ being decreasing,

$$\sum_{n \geq 2m} \frac{1}{n^5} \leq \int_{2m-1}^{\infty} \frac{dt}{t^5} = \left[\frac{-1}{4t^4} \right]_{2m-1}^{\infty} = \frac{1}{4(2m-1)^4} \leq \frac{1}{4m^4},$$

Zone 3 is therefore bounded above by

$$\frac{8}{7} m^3 \sum_{n \geq 2m} \frac{1}{n^5} \leq \frac{8}{7} m^3 \cdot \frac{1}{4m^4} = \frac{8}{7} \cdot \frac{1}{4m} = \frac{2}{7m} \leq 1.$$

(Zone 2): $\frac{m}{2} < n < 2m$ and $n \neq m$. We factor the difference of cubes:

$$\left| \frac{1}{n^3} - \frac{1}{m^3} \right| = \frac{|m^3 - n^3|}{n^3 m^3} = \frac{|m - n| (m^2 + mn + n^2)}{n^3 m^3}.$$

Since all the terms m^2 , mn , n^2 are positive, we bound $m^2 + mn + n^2 \geq m^2$ from below, whence

$$\left| \frac{1}{n^3} - \frac{1}{m^3} \right| \geq \frac{|m - n| m^2}{n^3 m^3} = \frac{|m - n|}{n^3 m}.$$

Carrying this over, each term of zone 2 is bounded above by

$$\frac{1}{n^5 |n^{-3} - m^{-3}|} \leq \frac{1}{n^5} \cdot \frac{n^3 m}{|m - n|} = \frac{m}{n^2 |m - n|}.$$

Now in this zone $n > \frac{m}{2}$, so $n^2 > \frac{m^2}{4}$ and $\frac{1}{n^2} < \frac{4}{m^2}$; thus

$$\frac{m}{n^2 |m - n|} < \frac{m}{\frac{m^2}{4} |m - n|} = \frac{4}{m |m - n|}.$$

Let us sum $\frac{4}{m |m - n|}$ over the n of the zone. I set $j = |m - n|$: as n runs through the integers from $\lceil \frac{m}{2} \rceil$ to $\lfloor 2m \rfloor$ while avoiding $n = m$, the integer j ranges between 1 and at most m , and each value of j is attained at most twice (once for $n = m - j$, once for $n = m + j$). Hence

$$\sum_{\text{Zone 2}} \frac{4}{m |m - n|} \leq \frac{4}{m} \cdot 2 \sum_{j=1}^m \frac{1}{j} = \frac{8}{m} H_m,$$

where $H_m = \sum_{j=1}^m \frac{1}{j}$ is the partial harmonic sum. Using the classical bound $H_m \leq 1 + \ln m$ (obtained by comparing the series with an integral), the contribution of zone 2 is bounded above by

$$\frac{8 H_m}{m} \leq \frac{8(1 + \ln m)}{m}.$$

The function $\frac{8(1 + \ln m)}{m}$ is decreasing for $m \geq 1$. Its maximum over the integers $m \geq 1$ is therefore attained at $m = 1$, where $\frac{8(1 + \ln 1)}{1} = 8$. Zone 2 is thus bounded above by 8 for every $m \geq 1$.

Summary. Adding up the three contributions:

$$\Sigma_m \leq \underbrace{2}_{\text{Zone 1}} + \underbrace{8}_{\text{Zone 2}} + \underbrace{1}_{\text{Zone 3}} = 11 =: C,$$

and this bound is indeed *independent of m* , which was the goal.

The term T_m being real and positive, we may apply the reverse triangle inequality, and we then obtain

$$|f(z_m)| = \left| T_m + \sum_{n \neq m} \frac{1}{n^5(\alpha_n - z_m)} \right| \geq T_m - \Sigma_m \geq m - C \xrightarrow{m \rightarrow \infty} +\infty.$$

The function f is therefore not bounded on the unit circle $\mathbb{U}$.

1.5.4 Conclusion

Each partial sum S_K is bounded on $\mathbb{D}$, since $|S_K(z)| \leq \sum_{k=0}^K |a_k| |z|^k \leq \sum_{k=0}^K |a_k| \leq 2(K+1)$.

Suppose, for contradiction, that the convergence were uniform on $\mathbb{U}$. A uniform limit of bounded functions is bounded (by Proposition 1.1): indeed, taking $\varepsilon = 1$ in the uniform convergence, there exists K_0 with $\sup_{\mathbb{U}} |f - S_{K_0}| \leq 1$, whence $|f| \leq 1 + \sup_{\mathbb{U}} |S_{K_0}| < \infty$ on $\mathbb{U}$. This contradicts the unboundedness of f established above.

Hence the convergence of $\sum_{k \geq 0} a_k z^k$ is uniform neither on $\mathbb{U}$, nor on $\mathbb{D}$ which contains it, even though the series converges at *every* point of $\mathbb{D}$.

Finally, the radius of convergence is exactly $R = 1$: we already have $R \geq 1$; and if we had $R > 1$, the series would converge normally on $\mathbb{D}$, which has just been ruled out.

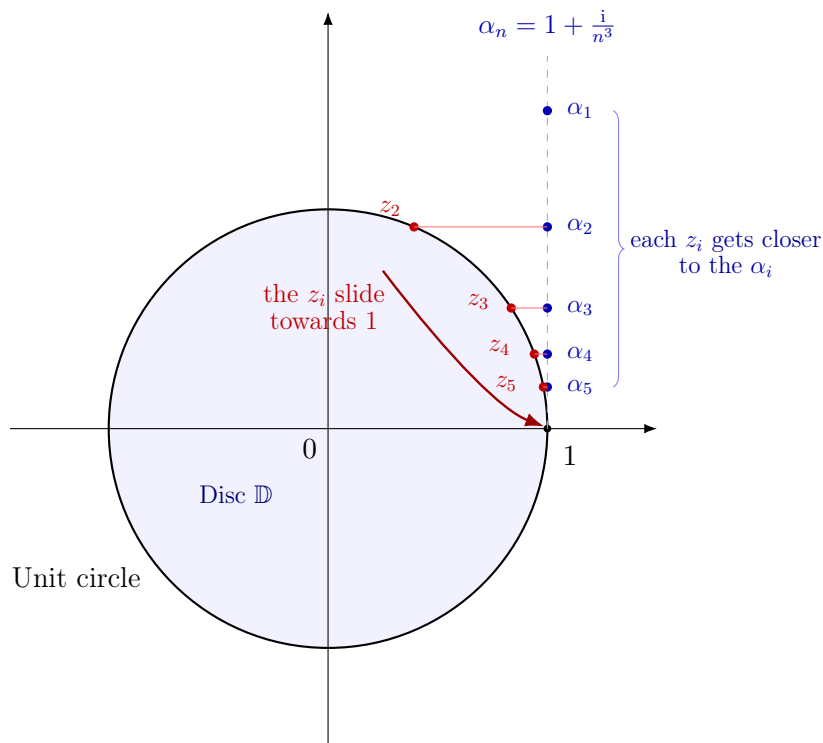


FIGURE 1.1: Domain of convergence of the series

1.6 One last question

The pair of exponents $(s, t) = (5, 3)$ used by Lowther may look arbitrary. The aim here is to show that there is no magic in the values 5 and 3, and to explain why they are nevertheless good pedagogical choices.

1.6.1 The conditions to be satisfied

Let us recall the criterion established above. Generalising the construction with $\alpha_n = 1 + \frac{i}{n^t}$ and a factor $\frac{1}{n^s}$, two conditions must be satisfied:

$$\underbrace{s > t + 1}_{\text{(I) convergence at } z=1} \quad \text{and} \quad \underbrace{s < 2t}_{\text{(II) unboundedness on } \mathbb{U}}.$$

Every pair (s, t) with $t > 1$ and

$$t + 1 < s < 2t$$

therefore yields a power series converging at every point of $\overline{\mathbb{D}}$ without converging uniformly there. The pair $(5, 3)$ satisfies $4 < 5 < 6$.

1.6.2 The set of admissible pairs

We can exhibit a set of pairs (s, t) that work:

t	interval $]t + 1, 2t[$	example of a working s
2	$]3, 4[$	$s = 3.5$
3	$]4, 6[$	$s = 5$
4	$]5, 8[$	$s = 6$ or 7
5	$]6, 10[$	$s = 7, 8$ or 9

We observe that the pair $(5, 3)$ is the simplest choice, since it gives us a pair of integers.

1.6.3 Conclusion

The choice $(5, 3)$ is one that simplifies the computations, for instance the evaluation of $f(1) = \zeta(2) = \frac{\pi^2}{6}$, and makes the estimates easy.

Chapter 2

Generating series

2.1 Introduction

In the previous chapter, we studied power series for their own sake, focusing on their analytic behaviour at the boundary of the disc of convergence. In this chapter, I will show that these same objects provide a remarkably effective probabilistic tool: the **generating function** of an integer-valued random variable.

The idea is as follows. Rather than handling a probability distribution $(p_n)_{n \geq 0}$ directly as a sequence of numbers, we encode it in a single function, namely the sum of the power series $\sum_{n \geq 0} p_n t^n$. This operation turns probabilistic questions (computing an expectation, a variance, the law of a sum of independent variables) into analytic manipulations (differentiation, product of series). It is precisely the regularity of power series inside the disc of convergence, established in Chapter 1, that makes these manipulations legitimate.

All the information about the law of X is contained in its generating function. The problem, as is often the case, is not to put the information in — we have put all of it

in — but to know how to get it out. This chapter sets out to develop that toolbox, from the first definitions up to the transfer theorem for a sum of a random number of random variables.

2.2 Preliminaries

Definition 2.1. Let $p \in [1, +\infty[$. A random variable X is said to be **of class $\mathcal{L}^p$** if $E(|X|^p) < +\infty$. We write $\mathcal{L}^p = \mathcal{L}^p(\Omega, \mathcal{C}, P)$ for the set of such random variables.

Definition 2.2. A random variable X is said to be **essentially bounded**, or of class $\mathcal{L}^\infty$, if there exists a constant $C > 0$ such that

$$\forall \omega \in \Omega, \quad |X(\omega)| \leq C$$

(in other words $\sup_{\omega \in \Omega} |X(\omega)| < +\infty$). The number $\|X\|_\infty = \sup_{\omega \in \Omega} |X(\omega)|$ is then called the $\mathcal{L}^\infty$ **norm** of the r.v. X .

Proposition 2.1. If $X \in \mathcal{L}^\infty$, then $X \in \mathcal{L}^p$ for every real $p > 0$.

Remark 2.1. On a *probability* space, one has more generally the decreasing inclusion $\mathcal{L}^q \subset \mathcal{L}^p$ as soon as $0 < p \leq q$. In particular $\mathcal{L}^2 \subset \mathcal{L}^1$, a fact that we shall use freely.

2.3 Generating functions

A method of very general scope consists in associating with every discrete r.v. X taking values in $\mathbb{N}$ the power series $G_X(t) = \sum_{n \geq 0} p_n t^n$, where $p_n = P(X = n)$.

We shall see that all the information concerning the law of X is contained in $G_X(t)$.

Definition 2.3. Let X be an r.v. taking values in $\mathbb{N}$, with law $p_n = P(X = n)$, $n \in \mathbb{N}$, where $\sum_{n \geq 0} p_n = 1$. The **generating function** of X is the power series

$$G_X(t) = \sum_{n \geq 0} p_n t^n = E(t^X).$$

Remark 2.2. G_X is a power series whose radius of convergence is *at least* 1. Indeed, since $0 \leq p_n \leq 1$, we have $\limsup_n |p_n|^{1/n} \leq 1$, so $R = 1/\limsup_n |p_n|^{1/n} \geq 1$. More elementarily, $G_X(1) = \sum_{n \geq 0} p_n = 1 < +\infty$, so the series converges (absolutely, since the p_n are non-negative) for every $|t| \leq 1$.

Remark 2.3. The notation $G_X(t) = E(t^X)$ is consistent: by the transfer theorem for a discrete r.v., $E(t^X) = \sum_{n \geq 0} t^n P(X = n) = \sum_{n \geq 0} p_n t^n$.

Proposition 2.2 (The generating function characterises the law). Two r.v. X and Y taking values in $\mathbb{N}$ have the same law if and only if $G_X = G_Y$ on a neighbourhood of 0 (or simply on $[0, 1]$).

Proof. The direct implication is immediate. Conversely, assume $G_X(t) = G_Y(t)$ for every t in a neighbourhood of 0. A power series with radius of convergence ≥ 1 determines its coefficients uniquely: by Theorem 1.3, differentiating n times and evaluating at $t = 0$,

$$p_n = P(X = n) = \frac{G_X^{(n)}(0)}{n!} = \frac{G_Y^{(n)}(0)}{n!} = P(Y = n).$$

The two laws therefore coincide term by term. □

2.3.1 Generating function of a sum of independent variables

Theorem 2.1. Let G_X and G_Y be the respective generating functions of two independent r.v. X and Y taking values in $\mathbb{N}$. Then the generating function G_{X+Y} of the r.v. $X + Y$ is given by the formula

$$G_{X+Y}(t) = G_X(t) G_Y(t).$$

Proof. Set $G_X(t) = \sum_{i \geq 0} a_i t^i$ and $G_Y(t) = \sum_{j \geq 0} b_j t^j$, with $a_i = P(X = i)$ and $b_j = P(Y = j)$. We compute the law of $X + Y$ by the law of total probability, conditioning on the value of Y . For every $k \geq 0$:

$$\begin{aligned} P(X + Y = k) &= \sum_j P(X + Y = k, Y = j) = \sum_{j=0}^k P(X = k - j, Y = j) \\ &= \sum_{j=0}^k P(X = k - j) P(Y = j) \quad (\text{since } X \text{ and } Y \text{ are independent}) \\ &= \sum_{j=0}^k a_{k-j} b_j. \end{aligned}$$

We recognise the Cauchy product of the two series: $\sum_{j=0}^k a_{k-j} b_j$ is exactly the coefficient of t^k in the product $G_X(t)G_Y(t)$. Since the two power series G_X and G_Y have radius of convergence at least 1, their Cauchy product converges and represents $G_X(t)G_Y(t)$ for $|t| < 1$, and even at $|t| = 1$ by absolute convergence. Hence, for every $|t| \leq 1$,

$$G_{X+Y}(t) = \sum_{k \geq 0} \left(\sum_{j=0}^k a_{k-j} b_j \right) t^k = G_X(t) G_Y(t). \quad \square$$

Remark 2.4. By an immediate induction, if $X_1, \dots, X_n$ are n independent r.v. taking values in $\mathbb{N}$, then

$$G_{X_1+\dots+X_n}(t) = \prod_{i=1}^n G_{X_i}(t).$$

In particular, if the X_i furthermore have the *same law* (i.i.d.), with common generating function G , then $G_{X_1+\dots+X_n}(t) = (G(t))^n$.

Application A box contains four balls numbered 0, 1, 1, 2. We perform n draws with replacement. Let S_n be the sum of the numbers drawn ($n \geq 1$). Determine the probability distribution of the r.v. S_n .

On a single draw, the r.v. S_1 takes the value 0 with probability $\frac{1}{4}$, the value 1 with probability $\frac{2}{4} = \frac{1}{2}$ (two balls carry the number 1) and the value 2 with probability $\frac{1}{4}$. Its generating function is therefore

$$G_{S_1}(t) = \frac{1}{4} + \frac{1}{2}t + \frac{1}{4}t^2 = \frac{1+2t+t^2}{4} = \left(\frac{1+t}{2}\right)^2.$$

Since the draws are made with replacement, we may regard S_n as the sum of n independent r.v. with the same law as S_1 . By Theorem 2.1 (i.i.d. version):

$$G_{S_n}(t) = (G_{S_1}(t))^n = \left(\frac{1+t}{2}\right)^{2n} = \left(\frac{1}{2}\right)^{2n} (1+t)^{2n} = \frac{1}{2^{2n}} \sum_{k=0}^{2n} \binom{2n}{k} t^k.$$

Identifying the coefficient of t^k , we deduce that for every $k = 0, 1, \dots, 2n$:

$$\boxed{P(S_n = k) = \frac{1}{2^{2n}} \binom{2n}{k}.$$

We check in passing that $\sum_{k=0}^{2n} P(S_n = k) = \frac{1}{2^{2n}} \sum_{k=0}^{2n} \binom{2n}{k} = \frac{2^{2n}}{2^{2n}} = 1$.

2.4 Computing moments by differentiation

The main interest of the generating function is that it lets us read off the moments of X from the derivatives of G at the point $t = 1$. The subtlety — and this is where all the work of Chapter 1 comes in — is that $t = 1$ lies on the *boundary* of the disc of convergence: the passage to the radial limit must be justified.

Theorem 2.2. Let X be a discrete r.v. taking values in $\mathbb{N}$ and let G be its generating function. Assume that $X \in \mathcal{L}^2$. Then the left-hand derivatives $G'(1)$ and $G''(1)$ of $G(t)$ exist at $t = 1$ and we have

$$\begin{aligned} E(X) &= G'(1), \\ E(X^2) &= G''(1) + G'(1), \\ \text{Var}(X) &= G''(1) + G'(1) - (G'(1))^2. \end{aligned}$$

Proof. A power series may always be differentiated formally inside its disc of convergence (Theorem 1.3), which gives, for every $t \in]-1, 1[$:

$$\begin{aligned} G'(t) &= \sum_{n \geq 1} n p_n t^{n-1}, \\ G''(t) &= \sum_{n \geq 2} n(n-1) p_n t^{n-2}. \end{aligned}$$

Consider the limit as $t \rightarrow 1^-$. The coefficients $n p_n$ and $n(n-1) p_n$ are non-negative. Since $X \in \mathcal{L}^2$ (hence $X \in \mathcal{L}^1$), the two series with non-negative terms

$$\sum_{n \geq 1} n p_n = E(X) < +\infty \quad \text{and} \quad \sum_{n \geq 2} n(n-1) p_n = E(X^2) - E(X) < +\infty$$

converge. By *Abel's theorem* (radial convergence, proved in Chapter 1), the functions G' and G'' extend continuously to $t = 1$, and their limiting value is the sum of the

series at $t = 1$. The left-hand derivatives $G'(1)$ and $G''(1)$ therefore make sense and satisfy

$$G'(1) = \sum_{n \geq 0} n p_n = E(X), \quad G''(1) = \sum_{n \geq 0} n(n-1) p_n = E(X^2) - E(X).$$

From this we get $E(X) = G'(1)$, then $E(X^2) = G''(1) + G'(1)$, and finally, by the König–Huygens formula:

$$\text{Var}(X) = E(X^2) - (E(X))^2 = G''(1) + G'(1) - (G'(1))^2. \quad \square$$

Application 2.1. Show that the generating function of a binomial r.v. X with parameters (n, p) is $G(t) = (pt + 1 - p)^n$. Deduce $E(X)$ and $\text{Var}(X)$.

Solution. Write $q = 1 - p$. By definition, $P(X = k) = \binom{n}{k} p^k q^{n-k}$ for $0 \leq k \leq n$. Newton's binomial formula gives directly

$$G(t) = \sum_{k=0}^n \binom{n}{k} p^k q^{n-k} t^k = \sum_{k=0}^n \binom{n}{k} (pt)^k q^{n-k} = (pt + q)^n = (pt + 1 - p)^n.$$

(Interpretation: a binomial $\mathcal{B}(n, p)$ is the sum of n i.i.d. Bernoulli r.v. with parameter p , whose generating function is $pt + q$; we recover $G(t) = (pt + q)^n$ by Theorem 2.1.)

Differentiating:

$$G'(t) = np(pt + q)^{n-1}, \quad G''(t) = n(n-1)p^2(pt + q)^{n-2}.$$

Since $p + q = 1$, we have $(p \cdot 1 + q)^m = 1$ for every m , so at $t = 1$:

$$G'(1) = np, \quad G''(1) = n(n-1)p^2.$$

Thus $E(X) = G'(1) = np$ and

$$\text{Var}(X) = G''(1) + G'(1) - G'(1)^2 = n(n-1)p^2 + np - (np)^2 = np - np^2 = np(1-p) = npq.$$

Application 2.2. Show that the generating function of a Poisson r.v. X with parameter $\lambda > 0$ is $G(t) = e^{\lambda(t-1)}$. Deduce $E(X)$ and $\text{Var}(X)$.

Solution. We have $P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$ for $k \geq 0$. Hence

$$G(t) = \sum_{k=0}^{\infty} e^{-\lambda} \frac{\lambda^k}{k!} t^k = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda t)^k}{k!} = e^{-\lambda} e^{\lambda t} = e^{\lambda(t-1)}.$$

This power series has infinite radius of convergence (it is the exponential), which allows us to differentiate and evaluate at $t = 1$ with no precautions about the boundary. We obtain $G'(t) = \lambda e^{\lambda(t-1)}$ and $G''(t) = \lambda^2 e^{\lambda(t-1)}$, whence $G'(1) = \lambda$ and $G''(1) = \lambda^2$. Consequently:

$$E(X) = \lambda, \quad \text{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda.$$

We recover the characteristic feature of the Poisson distribution: its expectation and its variance are both equal to λ .

Application 2.3. Let X and Y be two Poisson r.v. with respective parameters λ and $\mu > 0$. Assume X and Y independent. Show that $X + Y$ follows a Poisson distribution and give its parameter.

Solution. By Theorem 2.1 and Application 2.2:

$$G_{X+Y}(t) = G_X(t) G_Y(t) = e^{\lambda(t-1)} e^{\mu(t-1)} = e^{(\lambda+\mu)(t-1)}.$$

We recognise the generating function of a Poisson distribution with parameter $\lambda + \mu$. Since the generating function characterises the law (Proposition 2.2), we conclude that

$$X + Y \sim \mathcal{P}(\lambda + \mu).$$

(The Poisson distribution is therefore stable under independent summation, the parameters adding up.)

2.5 Random times and sums of a random number of variables

2.5.1 The notion of a random time

Let $(X_n)_{n \geq 0}$ be a sequence of independent r.v., which we shall think of as a process unfolding in time: X_n describes the state of the process at time n . For example, repeatedly and indefinitely tossing a coin is a process of independent r.v., and X_n denotes the outcome of the n -th toss. Let N be another r.v. taking values in $\mathbb{N}$, and assume that N and the sequence (X_n) are independent.

Definition 2.4. The state of the process at the random time N is the r.v. X_N defined by

$$X_N(\omega) = X_{N(\omega)}(\omega) \quad (\omega \in \Omega).$$

Example 2.1. We toss a coin and roll a die indefinitely, at the same time. Let X_n (resp. Y_n) be the outcome of the n -th toss for the coin (resp. of the n -th roll for the die). The X_n are independent r.v. with the same Bernoulli distribution of parameter p (the probability of obtaining “heads”, coded 1), and the Y_n are independent r.v. with the same uniform distribution on $\{1, \dots, 6\}$. The two sequences $(X_n)_{n \geq 0}$ and

$(Y_n)_{n \geq 0}$ are also independent of each other. Consider

$$N = \inf\{n \geq 0 ; X_n = 1\}$$

(the first time at which we obtain heads). The r.v. N is independent of the sequence $(Y_n)_{n \geq 0}$. The state of the sequence (Y_n) at the random time N is the r.v. Y_N , which describes the face shown by the die at the moment when we first obtain heads.

Application 2.4. With the assumptions of the beginning of this paragraph, assume in addition that the X_n are discrete and take values in the same set $\{x_k ; k \in \mathbb{N}\}$. Find the law of the r.v. X_N .

Solution. We apply the law of total probability, conditioning on the value of N :

$$\begin{aligned} P(X_N = x_k) &\stackrel{(1)}{=} \sum_{n=0}^{+\infty} P(X_N = x_k \mid N = n) P(N = n) \\ &\stackrel{(2)}{=} \sum_{n=0}^{\infty} P(X_n = x_k \mid N = n) P(N = n) \\ &\stackrel{(3)}{=} \sum_{n=0}^{\infty} P(X_n = x_k) P(N = n). \end{aligned}$$

Justifications: (1) law of total probability on the complete system of events $\{N = n\}_{n \geq 0}$; (2) on the event $\{N = n\}$ we have $X_N = X_n$; (3) X_n and N are independent.

Remark 2.5. If the X_n all have the same law, then $P(X_n = x_k) = P(X_0 = x_k)$ does not depend on n . In sum (3) we may therefore factor out $P(X_0 = x_k)$, and since $\sum_{n=0}^{\infty} P(N = n) = 1$, we find that X_N also has the same law as X_0 . *In other words, observing the process at an independent random time does not change the law, provided the process is identically distributed.*

Application 2.5. Let $p \in]0, 1[$. Assume that in the process $(X_n)_{n \geq 0}$, the r.v. X_n follows the binomial distribution with parameters (n, p) , and that the random time

N follows a Poisson distribution with parameter $\lambda > 0$. Show that X_N follows a Poisson distribution with parameter λp .

Solution. We compute the generating function of X_N by conditioning on N . Since $X_n \sim \mathcal{B}(n, p)$, we have $E(t^{X_n}) = (pt + 1 - p)^n$ by Application 2.1. By the independence of N and (X_n) :

$$\begin{aligned} G_{X_N}(t) &= E(t^{X_N}) = \sum_{n=0}^{\infty} E(t^{X_n} \mid N = n) P(N = n) = \sum_{n=0}^{\infty} (pt + 1 - p)^n e^{-\lambda} \frac{\lambda^n}{n!} \\ &= e^{-\lambda} \sum_{n=0}^{\infty} \frac{(\lambda(pt + 1 - p))^n}{n!} = e^{-\lambda} e^{\lambda(pt+1-p)} = e^{\lambda p(t-1)}. \end{aligned}$$

We recognise the generating function of a Poisson distribution with parameter λp (Application 2.2). By Proposition 2.2, $X_N \sim \mathcal{P}(\lambda p)$.

2.5.2 Sum of a random number of random variables

Let $(X_n)_{n \geq 1}$ be a sequence of r.v. with integer values ≥ 0 , independent and with the same generating function $G(t) = \sum_{n \geq 0} p_n t^n$. Let N be an r.v. with integer values ≥ 0 , independent of (X_n) and with generating function $H(t) = \sum_{n \geq 0} q_n t^n$. Consider the sum

$$S_N = X_1 + X_2 + \cdots + X_N,$$

with the convention $S_0 = 0$ (empty sum). This is an r.v. defined for every $\omega \in \Omega$ by $S_N(\omega) = X_1(\omega) + \cdots + X_{N(\omega)}(\omega)$. As before, we assume that the X_n and N are defined on the same probability space $(\Omega, \mathcal{C}, P)$.

Theorem 2.3 (Transfer / composition theorem). The r.v. S_N takes values in $\mathbb{N}$ and its generating function G_{S_N} is given by the composition formula

$$\underline{G_{S_N}(t) = H \circ G(t) = H(G(t))}.$$

Moreover, if the X_n and N are in $\mathcal{L}^2$, then $S_N \in \mathcal{L}^2$ and we have

$$\begin{aligned} E(S_N) &= E(N) E(X_1), \\ \text{Var}(S_N) &= E(N) \text{Var}(X_1) + (E(X_1))^2 \text{Var}(N). \end{aligned}$$

Proof. Step 1: the composition formula. For a fixed integer n , $S_n = X_1 + \cdots + X_n$ is a sum of n i.i.d. r.v. with generating function G , so by Theorem 2.1, $G_{S_n}(t) = (G(t))^n$. Write p_j^{*n} for the coefficient of t^j in $G_{S_n}(t)$, that is, $p_j^{*n} = P(S_n = j)$. Let $j \in \mathbb{N}$. By the law of total probability and the independence of N and the sequence (X_n) (we proceed as in Application 2.4):

$$P(S_N = j) = \sum_{n=0}^{\infty} P(X_1 + \cdots + X_n = j) P(N = n) = \sum_{n=0}^{\infty} p_j^{*n} q_n.$$

From this we deduce the generating function of S_N , interchanging the two summations (which is legitimate since all the terms are non-negative, by the Fubini–Tonelli theorem for double series):

$$\begin{aligned} G_{S_N}(t) &= \sum_{j=0}^{\infty} \left(\sum_{n=0}^{\infty} p_j^{*n} q_n \right) t^j = \sum_{n=0}^{\infty} q_n \left(\sum_{j=0}^{\infty} p_j^{*n} t^j \right) \\ &= \sum_{n=0}^{\infty} q_n G_{S_n}(t) \stackrel{(*)}{=} \sum_{n=0}^{\infty} q_n (G(t))^n = H(G(t)) = H \circ G(t), \end{aligned}$$

where $(*)$ uses $G_{S_n}(t) = (G(t))^n$, and where the last equality recognises the definition of H evaluated at the point $G(t)$: this is licit for $|t| \leq 1$, because then $|G(t)| \leq G(1) = 1$ stays inside the disc of convergence of H .

Step 2: expectation. We now assume $X_1, N \in \mathcal{L}^2$. The functions G and H are then twice left-differentiable at 1 (Theorem 2.2), so $H \circ G$ is as well. Differentiating the

composition by the chain rule:

$$(H \circ G)'(t) = H'(G(t)) G'(t).$$

At $t = 1$, since $G(1) = 1$:

$$E(S_N) = (H \circ G)'(1) = H'(1) G'(1) = E(N) E(X_1).$$

Step 3: variance. Differentiating a second time:

$$(H \circ G)''(t) = H''(G(t)) (G'(t))^2 + H'(G(t)) G''(t).$$

At $t = 1$ (with $G(1) = 1$):

$$(H \circ G)''(1) = H''(1) (G'(1))^2 + H'(1) G''(1).$$

By Theorem 2.2 applied to S_N :

$$\begin{aligned} \text{Var}(S_N) &= (H \circ G)''(1) + (H \circ G)'(1) - ((H \circ G)'(1))^2 \\ &= H''(1) G'(1)^2 + H'(1) G''(1) + H'(1) G'(1) - (H'(1) G'(1))^2. \end{aligned}$$

We now express each term by means of the moments. Write $m = E(X_1) = G'(1)$, $\sigma^2 = \text{Var}(X_1) = G''(1) + G'(1) - G'(1)^2$, and likewise $H'(1) = E(N)$, $H''(1) = E(N^2) - E(N) = E(N(N-1))$. Substituting $G''(1) = \sigma^2 - m + m^2$ and $H''(1) =$

$E(N^2) - E(N)$, then regrouping:

$$\begin{aligned} \text{Var}(S_N) &= (E(N^2) - E(N))m^2 + E(N)(\sigma^2 - m + m^2) + E(N)m - (E(N))^2 m^2 \\ &= E(N)\sigma^2 + m^2(E(N^2) - E(N) + E(N) - E(N)^2) \\ &= E(N)\text{Var}(X_1) + (E(X_1))^2 \text{Var}(N). \end{aligned}$$

That S_N is $\mathcal{L}^2$ follows from the finiteness of this variance. $\square$

2.6 Conclusion

At the end of this chapter, the generating function appears as a dictionary translating the probabilistic language into the analytic one. To a law on $\mathbb{N}$ there corresponds a power series of radius ≥ 1 ; to independence there corresponds the product (Theorem 2.1); to the moments there correspond the radial derivatives at 1 (Theorem 2.2); and to a sum of a random number of terms there corresponds the composition $H \circ G$ (Theorem 2.3).

The common thread with Chapter 1 is now clear: the whole computation of the moments rests on the possibility of extending the differentiated series radially up to the boundary of the disc, that is, on Abel's theorem. The point $t = 1$ becomes here the point at which all the probabilistic quantities can be read off. The boundary of the disc of convergence, far from being a mere analytic curiosity, is exactly where the statistical information lives.

Chapter 3

The Galton–Watson process

3.1 Introduction and statement of the problem

We now turn to a classical model of probability theory: the **Galton–Watson branching process**. The historical problem that gave rise to it is that of the extinction of family names. At the end of the 19th century, Francis Galton wondered about the gradual disappearance of surnames within the British aristocracy and put the question to Henry William Watson, who proposed a probabilistic model for it. The same scheme describes indifferently the transmission of a family name from father to son, the proliferation of particles in a chain reaction, the spread of an epidemic in its early stages, or the descent of a gene in a population: in each case, individuals independently produce a random number of offspring, following one and the same reproduction law.

3.2 Description of the model

Let us describe the model precisely. We consider particles that can give birth to particles of the same kind. We assume that there is one particle at the start, which constitutes generation zero. Each particle gives birth, independently of all the others, to a random number of particles: this number follows the same **reproduction law** $(p_k)_{k \geq 0}$, where p_k is the probability that a given particle produces exactly k offspring ($k = 0, 1, 2, \dots$). The offspring of all the particles of the n -th generation form the $(n + 1)$ -th generation, and the particles of each generation reproduce independently of one another.

Let Z_n denote the number of particles in the n -th generation. Thus $Z_0 = 1$, and if we write $X_{n,k}$ for the number of offspring produced by the k -th particle of the n -th generation, the dynamics of the process is given by the fundamental recurrence relation

$$Z_{n+1} = \sum_{k=1}^{Z_n} X_{n,k},$$

where the $X_{n,k}$ are independent r.v., all with the same reproduction law $(p_k)_{k \geq 0}$, and independent of Z_n . In this expression we recognise exactly the structure of a *sum of a random number of random variables* studied in the previous section: generation Z_{n+1} is a sum of Z_n i.i.d. terms.

The central question is that of **extinction**: with what probability does the line eventually die out, that is, does there exist an index n for which $Z_n = 0$? Intuitively, we expect the answer to depend on the mean number of offspring per particle, $m = E(X_{1,1}) = \sum_{k \geq 0} k p_k$. We shall see that the generating function turns this intuition into an exact and remarkably clean statement: the extinction probability is the smallest fixed point of the generating function G of the reproduction law, and it is the sign of $m - 1$ that decides the fate of the line.

3.2.1 Generating function of the generations and mean number of offspring

Write $G(t) = \sum_{k \geq 0} p_k t^k$ for the generating function of the reproduction law, and $G_n(t) = E(t^{Z_n})$ for the generating function of the n -th generation Z_n . Since $Z_0 = 1$ almost surely, we have $G_0(t) = t$, and $G_1 = G$. The recursive structure of the process then translates into an iteration of G .

Proposition 3.1. For every $n \geq 0$, the generating function of Z_n is the n -th iterate of G :

$$G_n = \underbrace{G \circ G \circ \cdots \circ G}_{n \text{ times}}, \quad \text{that is} \quad G_{n+1} = G_n \circ G = G \circ G_n.$$

Proof. We argue by induction on n .

Base case: for $n = 0$, $G_0 = \text{id}$ and the convention for the empty composition gives the result; the case $n = 1$ is the definition $G_1 = G$.

Induction step: assume the property holds at rank n , and let us show that it also holds at rank $n + 1$. The fundamental relation $Z_{n+1} = \sum_{k=1}^{Z_n} X_{n,k}$ expresses Z_{n+1} as a sum of a random number Z_n of i.i.d. r.v. $X_{n,k}$, with common generating function G , the number of terms Z_n being independent of the $X_{n,k}$ and having generating function G_n . The transfer theorem 2.3 therefore applies, with the “role of H ” played by G_n and the “role of G ” played by G :

$$G_{Z_{n+1}}(t) = G_n(G(t)) = (G_n \circ G)(t).$$

By the induction hypothesis $G_n = G^{\circ n}$, we obtain $G_{n+1} = G^{\circ n} \circ G = G^{\circ(n+1)}$. The commutation $G_n \circ G = G \circ G_n$ follows from the fact that an iterate commutes with itself. □

Let us now set

$$m = E(X_{1,1}) = \sum_{k \geq 0} k p_k = G'(1),$$

the **mean number of offspring** of a particle (assumed finite, i.e. $X_{1,1} \in \mathcal{L}^1$). The computation of the expectation of Z_n is immediate by differentiating the iterate.

Proposition 3.2. For every $n \geq 0$, we have $E(Z_n) = m^n$.

Proof. We argue by induction on n .

Base case. For $n = 0$, we have $Z_0 = 1$ almost surely, so $\mathbb{E}(Z_0) = 1 = m^0$. The property holds at rank 0.

Induction step. Assume $\mathbb{E}(Z_n) = m^n$ for some $n \geq 0$, and let us show $\mathbb{E}(Z_{n+1}) = m^{n+1}$. By Proposition 3.1, the generating functions satisfy $G_{n+1} = G_n \circ G$. Since G and G_n are power series of radius ≥ 1 , left-differentiable at 1, we differentiate to obtain

$$G'_{n+1}(t) = G'_n(G(t)) G'(t).$$

We evaluate at $t = 1$. Since $G(1) = \sum_k p_k = 1$, it follows that

$$G'_{n+1}(1) = G'_n(G(1)) G'(1) = G'_n(1) G'(1).$$

Now, by Theorem 2.2, the expectation of a random variable taking values in $\mathbb{N}$ can be read off from the derivative of its generating function at 1: $\mathbb{E}(Z_{n+1}) = G'_{n+1}(1)$, $\mathbb{E}(Z_n) = G'_n(1)$ and $m = G'(1)$. The previous relation therefore reads

$$\mathbb{E}(Z_{n+1}) = \mathbb{E}(Z_n) m.$$

Using the induction hypothesis $\mathbb{E}(Z_n) = m^n$, we conclude

$$\mathbb{E}(Z_{n+1}) = m^n \cdot m = m^{n+1}.$$

The property therefore holds at rank $n + 1$, which completes the induction: $\mathbb{E}(Z_n) = m^n$ for every $n \geq 0$. $\square$

This formula already gives a strong intuition: on average, the population is multiplied by m at each generation. If $m < 1$, the expectation $m^n \rightarrow 0$ and we expect extinction; if $m > 1$, the expectation $m^n \rightarrow +\infty$ and the population may explode.

3.2.2 The extinction probability

The event “the line dies out” is the increasing union of the events $\{Z_n = 0\}$: once a generation is empty, all the following ones are too. Set

$$q_n = P(Z_n = 0) = G_n(0), \quad \text{and} \quad \eta = P(\text{extinction}) = P\left(\bigcup_{n \geq 1} \{Z_n = 0\}\right).$$

Since the sequence of events $\{Z_n = 0\}$ is increasing (if $Z_n = 0$ then $Z_{n+1} = 0$), the continuity from below of the probability gives $\eta = \lim_{n \rightarrow \infty} q_n$.

Lemma 3.1. The sequence $(q_n)_{n \geq 0}$ satisfies $q_0 = 0$ and $q_{n+1} = G(q_n)$ for every $n \geq 0$. It is increasing and converges to a limit $\eta \in [0, 1]$.

Proof. We have $q_0 = P(Z_0 = 0) = 0$ since $Z_0 = 1$. By Proposition 3.1, $q_n = G_n(0) = G^{\circ n}(0)$, so

$$q_{n+1} = G_{n+1}(0) = G(G_n(0)) = G(q_n).$$

The function G is increasing on $[0, 1]$ (its coefficients p_k are non-negative). Since $q_1 = G(0) = p_0 \geq 0 = q_0$, an immediate induction using the monotonicity of G gives $q_{n+1} = G(q_n) \geq G(q_{n-1}) = q_n$: the sequence is increasing. Bounded above by 1, it converges to a limit $\eta \in [0, 1]$. $\square$

Theorem 3.1 (Characterisation of the extinction probability). Assume $p_1 \neq 1$ (otherwise the process is deterministic and trivial). The extinction probability η is the **smallest fixed point** of G in $[0, 1]$, that is, the smallest solution in $[0, 1]$ of the equation

$$G(s) = s.$$

Proof. η is a fixed point. The function G is continuous on $[0, 1]$ (the sum of a power series of radius ≥ 1 , continuous up to the boundary by the non-negativity of the coefficients). Passing to the limit in the relation $q_{n+1} = G(q_n)$ and using the continuity of G :

$$\eta = \lim_{n \rightarrow +\infty} q_{n+1} = \lim_{n \rightarrow +\infty} G(q_n) = G\left(\lim_n q_n\right) = G(\eta).$$

So η is a fixed point of G .

η is the smallest one. Let $s \in [0, 1]$ be any fixed point of G , i.e. $G(s) = s$. Let us show by induction that $q_n \leq s$ for every n . We have $q_0 = 0 \leq s$. If $q_n \leq s$, the monotonicity of G gives $q_{n+1} = G(q_n) \leq G(s) = s$. Hence $q_n \leq s$ for every n , and passing to the limit, $\eta \leq s$. The extinction probability is therefore less than or equal to every fixed point: it is the smallest one. $\square$

3.2.3 The extinction criterion: the role of the mean number

m

It remains to locate this smallest fixed point. Everything rests on the convexity of G and on the value of its slope $m = G'(1)$ at the point $t = 1$, which is itself always a fixed point since $G(1) = \sum_k p_k = 1$.

Theorem 3.2 (Galton–Watson extinction criterion). Assume $p_1 \neq 1$ and $X_{1,1} \in \mathcal{L}^1$, and set $m = G'(1) = E(X_{1,1})$.

1. **Critical case** ($m \leq 1$): extinction is almost sure, $\eta = 1$.
2. **Non-critical case** ($m > 1$): the extinction probability η is the unique fixed point of G in $[0, 1[$, and we have $0 < \eta < 1$; the line therefore survives with strictly positive probability $1 - \eta > 0$.

Proof. We set aside the degenerate case $p_0 = 0$: then $G(0) = 0$, extinction is impossible from the very first generation, but we also have $m = \sum_k k p_k \geq \sum_k p_k = 1$, with $m = 1$ only if $p_1 = 1$, which is excluded; so $p_0 = 0$ falls under the case $m > 1$ with $\eta = 0$, consistent with the statement. We now assume $p_0 > 0$, so that $G(0) = p_0 > 0$.

Let us study G on $[0, 1]$. Since $p_1 \neq 1$ and $p_0 > 0$, the law charges at least one integer $k \geq 2$, or else G is affine; in all the non-trivial cases, G is **convex** on $[0, 1]$ (because $G''(t) = \sum_{k \geq 2} k(k-1)p_k t^{k-2} \geq 0$), and even strictly convex as soon as $p_k > 0$ for some $k \geq 2$. Moreover $G(0) = p_0 > 0$, $G(1) = 1$, and G is increasing.

Consider the function $\varphi(s) = G(s) - s$ on $[0, 1]$. It is convex (the difference of a convex function and an affine one), and satisfies $\varphi(0) = p_0 > 0$ and $\varphi(1) = 0$. Its left-hand derivative at 1 is $\varphi'(1) = G'(1) - 1 = m - 1$.

Case $m \leq 1$. Then $\varphi'(1) = m - 1 \leq 0$. Since φ is convex, its derivative φ' is increasing; having $\varphi'(1) \leq 0$, we get $\varphi'(s) \leq 0$ for every $s \in [0, 1]$, so φ is decreasing on $[0, 1]$. Starting from $\varphi(0) = p_0 > 0$ and decreasing down to $\varphi(1) = 0$, the function φ vanishes *only* at $s = 1$ on $[0, 1]$ (it stays strictly positive on $[0, 1[$). The unique fixed point of G in $[0, 1]$ is therefore $s = 1$; by Theorem 3.1, the smallest fixed point is $\eta = 1$. Extinction is almost sure.

Case $m > 1$. Then $\varphi'(1) = m - 1 > 0$: the function φ is strictly increasing in a left neighbourhood of 1, so there exists $s_0 < 1$ with $\varphi(s_0) < \varphi(1) = 0$. But $\varphi(0) = p_0 > 0$. By the intermediate value theorem applied to φ , continuous on $[0, s_0]$, there exists $\eta \in]0, s_0[\subset]0, 1[$ such that $\varphi(\eta) = 0$, i.e. $G(\eta) = \eta$. This fixed point is unique in $[0, 1[$: indeed, a convex φ cannot vanish at more than two points of $[0, 1]$, and since $\varphi(1) = 0$ is already one of them, at most one zero remains in $[0, 1[$. The smallest fixed point of G is therefore this $\eta \in]0, 1[$, which is indeed < 1 . By Theorem 3.1, $\eta = P(\text{extinction})$ satisfies $0 < \eta < 1$, and survival occurs with probability $1 - \eta > 0$. $\square$

Remark 3.1 (Geometric interpretation). The criterion can be read off entirely from the graph of G compared with the diagonal $y = s$. The curve $y = G(s)$ starts at $G(0) = p_0 > 0$, is increasing and convex, and meets the diagonal at $s = 1$. Its slope at that point is m :

- if $m \leq 1$, the curve *comes down onto* the diagonal “from above” and has never crossed it before: the only fixed point is 1, extinction is certain;
- if $m > 1$, the curve *crosses* the diagonal earlier, at a point $\eta < 1$: this is where the extinction probability sits, strictly between 0 and 1.

The mean number of offspring m , that is, the slope of G at the boundary of the disc, thus decides on its own between the two regimes. Once again we recover the guiding

principle of this report: it is the behaviour of the power series *at the point* $t = 1$, on the boundary of its disc of convergence, that carries the decisive information.

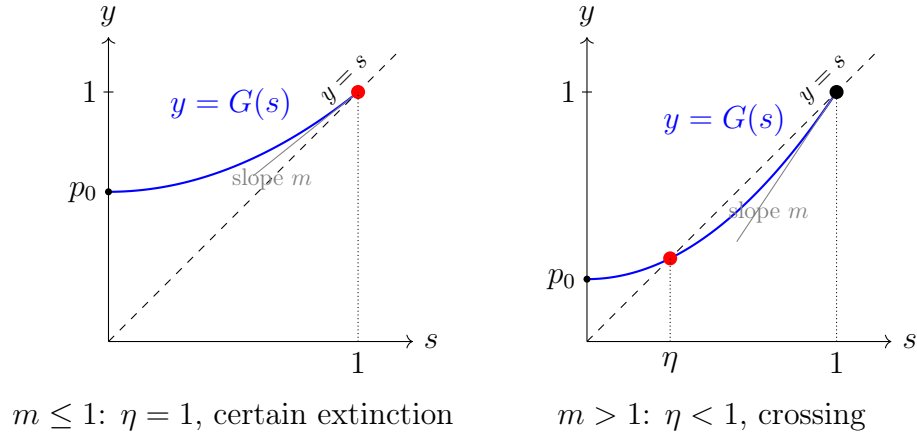


FIGURE 3.1: Graph of the generating function G compared with $y = s$

3.3 Applications

3.3.1 Bernoulli distribution

Suppose that the number of offspring X of a population follows a Bernoulli distribution with parameter p , that is, $P(X = 0) = q$ and $P(X = 1) = p$, where $q = 1 - p$.

Generating function. For $|s| \leq 1$:

$$G_X(s) = \sum_{k=0}^{\infty} s^k P(X = k) = q + ps$$

Fixed points. We solve $G_X(s) = s$:

$$q + ps = s \iff q + (p - 1)s = 0 \iff s = -\frac{q}{p - 1} \iff s = \frac{p - 1}{p - 1} = 1$$

We therefore have a unique fixed point (which is consistent, since we are dealing with a polynomial of degree 1).

Mean number of offspring. We differentiate:

$$G'_X(s) = (q + ps)' = p \quad \text{whence} \quad m = G'_X(1) = p$$

Extinction criterion. By Theorem 3.2, extinction is almost sure if $m \leq 1$:

$$m \leq 1 \iff p \leq 1$$

Thus:

- if $p < 1$ (hence $m < 1$), the line dies out almost surely, $\eta = 1$;
- if $p = 1$ (degenerate case), the line reproduces indefinitely.

3.3.2 Binomial distribution with $n = 2$

Suppose that the number of offspring X of a particle follows a binomial distribution with parameters $(2, p)$, that is, $P(X = k) = \binom{2}{k} p^k q^{2-k}$ for $k \in \{0, 1, 2\}$, where $q = 1 - p$.

Generating function. For every s :

$$G_X(s) = \sum_{k=0}^2 s^k \binom{2}{k} p^k q^{2-k} = q^2 + 2pq s + p^2 s^2 \underset{\text{binom. id.}}{=} (q + ps)^2.$$

Fixed points. We solve $G_X(s) = s$:

$$\begin{aligned}(q + ps)^2 = s &\iff q^2 + 2pq s + p^2 s^2 = s \\ &\iff p^2 s^2 + (2pq - 1)s + q^2 = 0.\end{aligned}$$

The real number 1 is always a fixed point (since $G_X(1) = 1$), hence an obvious root.

Using the identity $(p + q)^2 = 1$, i.e. $2pq - 1 = -(p^2 + q^2)$, the quadratic factors as

$$p^2 s^2 + (2pq - 1)s + q^2 = p^2(s - 1)\left(s - \frac{q^2}{p^2}\right),$$

which gives $s = \left(\frac{q}{p}\right)^2$ as the second root.

Mean number of offspring. We differentiate $G_X(s) = (q + ps)^2$:

$$G'_X(s) = 2p(q + ps) \quad \text{whence} \quad m = G'_X(1) = 2p(q + p) = 2p,$$

Extinction criterion. By Theorem 3.2, extinction is almost sure if and only if $m \leq 1$:

$$m \leq 1 \iff 2p \leq 1 \iff p \leq \frac{1}{2}.$$

Thus:

- if $p \leq \frac{1}{2}$ (hence $m \leq 1$), the line dies out almost surely, $\eta = 1$;
- if $p > \frac{1}{2}$ (hence $m > 1$), the second root $\left(\frac{q}{p}\right)^2 < 1$ becomes the smallest fixed point in $[0, 1]$, and it is this one that gives the extinction probability:

$$\eta = \left(\frac{q}{p}\right)^2 = \left(\frac{1-p}{p}\right)^2 \in]0, 1[.$$

3.3.3 Geometric distribution

Extinction probability Suppose that the number of offspring X of a population follows a geometric distribution with parameter p taking values in $\mathbb{N}$, that is, $P(X = k) = p q^k$ for $k \geq 0$, where $q = 1 - p$.

Generating function. For $|s| < 1/q$:

$$G_X(s) = \sum_{k=0}^{\infty} s^k p q^k = p \sum_{k=0}^{\infty} (qs)^k = \frac{p}{1 - qs}.$$

Fixed points. We solve $G_X(s) = s$:

$$\frac{p}{1 - qs} = s \iff p = s(1 - qs) \iff qs^2 - s + p = 0.$$

The real number 1 is always a fixed point (since $G_X(1) = 1$), hence an obvious root.

We may therefore write

$$qs^2 - s + p = q(s - 1)\left(s - \frac{p}{q}\right),$$

which gives $s = \frac{p}{q}$ as the second root.

Mean number of offspring. We differentiate:

$$G'_X(s) = \left(\frac{p}{1 - qs}\right)' = \frac{pq}{(1 - qs)^2}, \quad \text{whence} \quad m = G'_X(1) = \frac{pq}{(1 - q)^2} = \frac{pq}{p^2} = \frac{q}{p}.$$

Extinction criterion. By Theorem 3.2, extinction is almost sure if and only if $m \leq 1$:

$$m = \frac{q}{p} \leq 1 \iff q \leq p \iff 1 - p \leq p \iff \frac{1}{2} \leq p.$$

Thus:

- if $p \geq \frac{1}{2}$ (i.e. $m \leq 1$), the line dies out almost surely, $\eta = 1$;
- if $p < \frac{1}{2}$ (i.e. $m > 1$), we have $\eta < 1$ and the line survives with strictly positive probability. The extinction probability is then the smallest fixed point in $[0, 1[$, namely

$$\eta = \frac{p}{q} = \frac{p}{1-p} \in]0, 1[.$$

The two approaches agree: for $p \geq \frac{1}{2}$ we have $\frac{p}{q} \geq 1$, so the smallest fixed point in $[0, 1]$ is indeed 1; for $p < \frac{1}{2}$ we have $\frac{p}{q} < 1$, and it is this root that gives the extinction probability.

Study of the n -th generation By Proposition 3.1, the generating function of the n -th generation is the n -th iterate $G_n = G^{on}$. Now G is a **Möbius transformation** (a homography), and it is known that composing homographies corresponds to multiplying the matrices that represent them: to $G(s) = \frac{as+b}{cs+d}$ we associate the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, and then G^{on} is represented by the n -th power of that matrix. We may therefore diagonalise the matrix in order to obtain our n -th iterate.

The associated matrix. We write

$$G(s) = \frac{p}{1-qs} = \frac{0 \cdot s + p}{-q \cdot s + 1}, \quad \text{whence} \quad M = \begin{pmatrix} 0 & p \\ -q & 1 \end{pmatrix}.$$

Eigenvalues. The characteristic polynomial of M is

$$\chi_M(\lambda) = \lambda^2 - \lambda + pq,$$

with discriminant $\Delta = 1 - 4pq$. Assuming $\Delta \neq 0$, M has two distinct eigenvalues

$$\lambda_1 = \frac{1 + \sqrt{1 - 4pq}}{2}, \quad \lambda_2 = \frac{1 - \sqrt{1 - 4pq}}{2},$$

which satisfy the relations $\lambda_1 + \lambda_2 = 1$ and $\lambda_1\lambda_2 = pq$.

Change-of-basis matrix and its inverse. We therefore set

$$P = \begin{pmatrix} p & p \\ \lambda_1 & \lambda_2 \end{pmatrix}, \quad \det P = p(\lambda_2 - \lambda_1) = -p\sqrt{\Delta}, \quad P^{-1} = \frac{1}{p(\lambda_2 - \lambda_1)} \begin{pmatrix} \lambda_2 & -p \\ -\lambda_1 & p \end{pmatrix}.$$

The n -th power: the product PD^nP^{-1} . With $D = \text{diag}(\lambda_1, \lambda_2)$, we have $M^n = PD^nP^{-1}$, that is, the product of the three matrices

$$M^n = \begin{pmatrix} p & p \\ \lambda_1 & \lambda_2 \end{pmatrix} \begin{pmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{pmatrix} \cdot \frac{1}{p(\lambda_2 - \lambda_1)} \begin{pmatrix} \lambda_2 & -p \\ -\lambda_1 & p \end{pmatrix}.$$

We first carry out the product of the first two matrices:

$$\begin{pmatrix} p & p \\ \lambda_1 & \lambda_2 \end{pmatrix} \begin{pmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{pmatrix} = \begin{pmatrix} p\lambda_1^n & p\lambda_2^n \\ \lambda_1^{n+1} & \lambda_2^{n+1} \end{pmatrix},$$

then we multiply by P^{-1} . Pulling the factor $\frac{1}{p(\lambda_2 - \lambda_1)} = \frac{-1}{p(\lambda_1 - \lambda_2)}$ out in front, we obtain

$$M^n = \frac{-1}{p(\lambda_1 - \lambda_2)} \begin{pmatrix} p\lambda_1^n & p\lambda_2^n \\ \lambda_1^{n+1} & \lambda_2^{n+1} \end{pmatrix} \begin{pmatrix} \lambda_2 & -p \\ -\lambda_1 & p \end{pmatrix}.$$

The product of the last two matrices gives, entry by entry,

$$\begin{pmatrix} p\lambda_1^n\lambda_2 - p\lambda_2^n\lambda_1 & -p^2\lambda_1^n + p^2\lambda_2^n \\ \lambda_1^{n+1}\lambda_2 - \lambda_2^{n+1}\lambda_1 & -p\lambda_1^{n+1} + p\lambda_2^{n+1} \end{pmatrix}.$$

Distributing the prefactor $\frac{-1}{p(\lambda_1 - \lambda_2)}$ and simplifying (using $\lambda_1\lambda_2 = pq$ for the bottom-left entry: $\lambda_1^{n+1}\lambda_2 - \lambda_2^{n+1}\lambda_1 = \lambda_1\lambda_2(\lambda_1^n - \lambda_2^n) = pq(\lambda_1^n - \lambda_2^n)$), we finally get

$$M^n = \frac{1}{\lambda_1 - \lambda_2} \begin{pmatrix} \lambda_1\lambda_2^n - \lambda_1^n\lambda_2 & p(\lambda_1^n - \lambda_2^n) \\ -q(\lambda_1^n - \lambda_2^n) & \lambda_1^{n+1} - \lambda_2^{n+1} \end{pmatrix}.$$

Verification by induction. Let us check that this expression does satisfy $M^{n+1} = M^n \cdot M$. We multiply the boxed formula by $M = \begin{pmatrix} 0 & p \\ -q & 1 \end{pmatrix}$, keeping the factor $\frac{1}{\lambda_1 - \lambda_2}$ outside:

$$M^n \cdot M = \frac{1}{\lambda_1 - \lambda_2} \begin{pmatrix} \lambda_1\lambda_2^n - \lambda_1^n\lambda_2 & p(\lambda_1^n - \lambda_2^n) \\ -q(\lambda_1^n - \lambda_2^n) & \lambda_1^{n+1} - \lambda_2^{n+1} \end{pmatrix} \begin{pmatrix} 0 & p \\ -q & 1 \end{pmatrix}.$$

Let us compute the four entries of the product (always up to the factor $\frac{1}{\lambda_1 - \lambda_2}$).

Top-left corner. $(\lambda_1\lambda_2^n - \lambda_1^n\lambda_2) \cdot 0 + p(\lambda_1^n - \lambda_2^n) \cdot (-q) = -pq(\lambda_1^n - \lambda_2^n)$.

Now $-pq = -\lambda_1\lambda_2$, so this term equals $\lambda_1\lambda_2(\lambda_2^n - \lambda_1^n) = \lambda_1\lambda_2^{n+1} - \lambda_1^{n+1}\lambda_2$, which is indeed the top-left corner of M^{n+1} .

Top-right corner. $(\lambda_1\lambda_2^n - \lambda_1^n\lambda_2) \cdot p + p(\lambda_1^n - \lambda_2^n) \cdot 1 = p(\lambda_1\lambda_2^n - \lambda_1^n\lambda_2 + \lambda_1^n - \lambda_2^n)$.

Regrouping, $\lambda_1^n(1 - \lambda_2) - \lambda_2^n(1 - \lambda_1) = \lambda_1^n\lambda_1 - \lambda_2^n\lambda_2 = \lambda_1^{n+1} - \lambda_2^{n+1}$ (we have used

$1 - \lambda_2 = \lambda_1$ and $1 - \lambda_1 = \lambda_2$, that is, $\lambda_1 + \lambda_2 = 1$). This term therefore equals $p(\lambda_1^{n+1} - \lambda_2^{n+1})$, the top-right corner of M^{n+1} .

Bottom-left corner. $-q(\lambda_1^n - \lambda_2^n) \cdot 0 + (\lambda_1^{n+1} - \lambda_2^{n+1}) \cdot (-q) = -q(\lambda_1^{n+1} - \lambda_2^{n+1})$, the bottom-left corner of M^{n+1} .

Bottom-right corner. $-q(\lambda_1^n - \lambda_2^n) \cdot p + (\lambda_1^{n+1} - \lambda_2^{n+1}) \cdot 1 = (\lambda_1^{n+1} - \lambda_2^{n+1}) - pq(\lambda_1^n - \lambda_2^n)$. Each eigenvalue satisfies $\lambda_i^2 = \lambda_i - pq$, so multiplying by λ_i^n : $\lambda_i^{n+2} = \lambda_i^{n+1} - pq \lambda_i^n$. Subtracting the two relations ($i = 1, 2$), we obtain exactly $(\lambda_1^{n+1} - \lambda_2^{n+1}) - pq(\lambda_1^n - \lambda_2^n) = \lambda_1^{n+2} - \lambda_2^{n+2}$. This term therefore equals $\lambda_1^{n+2} - \lambda_2^{n+2}$, the bottom-right corner of M^{n+1} .

All four corners agree: we do have $M^n \cdot M = M^{n+1}$. The boxed formula, true at rank $n = 1$ (we recover M), is therefore established for every $n \geq 1$.

Generating function of the n -th generation. Reading off the homography associated with $M^n = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$, namely $s \mapsto \frac{As + B}{Cs + D}$ (the common factor $\frac{1}{\lambda_1 - \lambda_2}$ cancels out), we obtain

$$G_n(s) = G^{on}(s) = \frac{(\lambda_1 \lambda_2^n - \lambda_1^n \lambda_2) s + p(\lambda_1^n - \lambda_2^n)}{-q(\lambda_1^n - \lambda_2^n) s + (\lambda_1^{n+1} - \lambda_2^{n+1})}$$

3.3.4 An application in biology: replication of a nucleotide strand

First of all, a nucleotide is the basic building block making up nucleic acids, that is, DNA and RNA, or the molecules that carry genetic information. A nucleotide strand is a group of nucleotides.

Consider a strand made up of l nucleotides, subject to the following reproduction law. At each generation, the strand survives with probability $\omega \in]0, 1]$ (and disappears

with probability $1 - \omega$). If it survives, each of its l positions replicates faithfully with probability p , independently of the others; the copy produced is therefore exact with probability p^l , in which case the strand produces 2 identical offspring, and imperfect with probability $1 - p^l$, in which case it produces only one. The reproduction law of the number X of offspring is thus

$$P(X = 0) = 1 - \omega, \quad P(X = 1) = \omega(1 - p^l), \quad P(X = 2) = \omega p^l.$$

Generating function. For every s :

$$G_X(s) = \sum_{k=0}^2 s^k P(X = k) = (1 - \omega) + \omega(1 - p^l)s + \omega p^l s^2.$$

Fixed points. We solve $G_X(s) = s$:

$$(1 - \omega) + \omega(1 - p^l)s + \omega p^l s^2 = s \iff \omega p^l s^2 + (\omega(1 - p^l) - 1)s + (1 - \omega) = 0.$$

The real number 1 is an obvious root (since $G_X(1) = 1$). The leading coefficient being ωp^l , we factor:

$$\omega p^l s^2 + (\omega(1 - p^l) - 1)s + (1 - \omega) = \omega p^l (s - 1) \left(s - \frac{1 - \omega}{\omega p^l} \right),$$

which gives $s = \frac{1 - \omega}{\omega p^l}$ as the second root.

Mean number of offspring. We differentiate:

$$G'_X(s) = \omega(1 - p^l) + 2\omega p^l s \quad \text{whence} \quad m = G'_X(1) = \omega(1 - p^l) + 2\omega p^l = \omega(1 + p^l).$$

Extinction criterion. By Theorem 3.2, extinction is almost sure if and only if $m \leq 1$.

Since $t \mapsto t^{1/l}$ is increasing:

$$m \leq 1 \iff \omega(1 + p^l) \leq 1 \iff p^l \leq \frac{1}{\omega} - 1 \iff p \leq \left(\frac{1}{\omega} - 1\right)^{1/l}.$$

Write $p^* = \left(\frac{1}{\omega} - 1\right)^{1/l}$ for the critical threshold. Thus:

- if $p \leq p^*$ (hence $m \leq 1$), the strand disappears almost surely, $\eta = 1$;
- if $p > p^*$ (hence $m > 1$), the line has a strictly positive probability of perpetuating itself indefinitely; the extinction probability is then the smallest fixed point, namely

$$\eta = \frac{1 - \omega}{\omega p^l} \in]0, 1[.$$

Interpretation: the higher the replication fidelity p (beyond the threshold p^*), the better the strand's chances of proliferating without end; below it, the accumulation of replication failures condemns the line to extinction.

Chapter 4

Random walk

4.1 Introduction

The *simple random walk* on $\mathbb{Z}$ is one of the most elementary objects of probability theory, and yet one of the richest. It models the motion of a particle which, at each integer time, takes a step upwards with probability $p \in]0, 1[$ or downwards with probability $q = 1 - p$, independently of its past.

Formally, we are given a sequence $(\xi_n)_{n \geq 1}$ of independent and identically distributed random variables, taking values in $\{-1, +1\}$, such that

$$\mathbb{P}(\xi_n = +1) = p, \quad \mathbb{P}(\xi_n = -1) = q = 1 - p.$$

The random walk started at the origin is then defined by

$$S_0 = 0, \quad S_n = \sum_{k=1}^n \xi_k \quad (n \geq 1).$$

The sequence $(S_n)_{n \geq 0}$ describes the position of the particle over time. When $p = q = \frac{1}{2}$, the walk is called *symmetric*; otherwise the walk has a *drift* of expectation $\mathbb{E}[\xi_1] = p - q$, which orients the mean displacement towards $+\infty$ or $-\infty$.

Despite the simplicity of its definition, the random walk raises deep questions. Among the most natural ones are those of the *return to the origin* and of the *first passage times*: does the particle almost surely return to 0? After how many steps does it reach a given level for the first time? We shall see that these questions admit precise answers, involving a variety of tools.

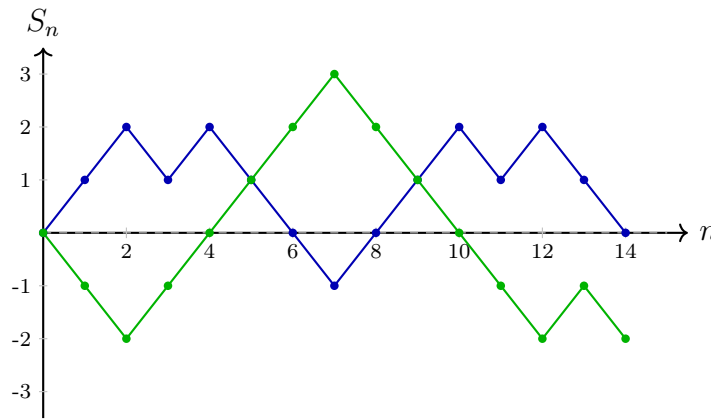


FIGURE 4.1: Examples of random walks

4.2 Return to 0

We are interested in the probability that the walk returns to the origin at a given time. Recall that the particle goes up (makes a $+1$ step) with probability p and goes down (makes a -1 step) with probability $q = 1 - p$, independently at each step.

We look for the probability $P(S_n = 0)$ that the walk has returned to 0 at time n . Let us first observe a *parity* constraint: at each step, the position S_n changes parity. Starting from $S_0 = 0$ (even), the walk can therefore only be back at 0 at *even* times. We thus write $n = 2m$ and study $P(S_{2m} = 0)$.

To be back at 0 after $2m$ steps, the path must contain exactly m up-steps and m down-steps (so that the net displacement $m - m$ is zero). A fixed path satisfying this condition has probability

$$\underbrace{p \cdots p}_m \cdot \underbrace{q \cdots q}_m = p^m q^m,$$

by the independence of the steps. It remains to count these paths: choosing a path amounts to choosing, among the $2m$ steps, the m that are up-steps, which gives $\binom{2m}{m}$ possibilities. We finally obtain

$$P(S_{2m} = 0) = \binom{2m}{m} p^m q^m.$$

Stirling's formula and the search for an equivalent. Stirling's formula gives the equivalent $n! \underset{\infty}{\sim} \sqrt{2\pi n} n^n e^{-n}$. From this we deduce

$$\begin{aligned} (n!)^2 &\underset{\infty}{\sim} (\sqrt{2\pi n} n^n e^{-n})^2 = 2\pi n n^{2n} e^{-2n}, \\ (2n)! &\underset{\infty}{\sim} \sqrt{4\pi n} (2n)^{2n} e^{-2n} = \sqrt{4\pi n} 2^{2n} n^{2n} e^{-2n}, \end{aligned}$$

whence, simplifying the factors $n^{2n} e^{-2n}$ and noting that $\frac{\sqrt{4\pi n}}{2\pi n} = \frac{1}{\sqrt{\pi n}}$,

$$\binom{2n}{n} = \frac{(2n)!}{(n!)^2} \underset{\infty}{\sim} \frac{\sqrt{4\pi n} 2^{2n} n^{2n} e^{-2n}}{2\pi n n^{2n} e^{-2n}} = \frac{2^{2n}}{\sqrt{\pi n}} = \frac{4^n}{\sqrt{\pi n}}.$$

From this we deduce the equivalent of the return probability:

$$P(S_{2n} = 0) = \binom{2n}{n} p^n q^n \underset{\infty}{\sim} \frac{(4pq)^n}{\sqrt{\pi n}}.$$

The behaviour of the series $\sum_n P(S_{2n} = 0)$ then depends on the product $4pq$. Since

$p + q = 1$, we have $4pq = 4p(1 - p) \leq 1$, with equality if and only if $p = q = \frac{1}{2}$. Two cases arise.

- **Symmetric case** ($p = q = \frac{1}{2}$). Then $4pq = 1$ and

$$P(S_{2n} = 0) \sim \frac{1}{\sqrt{\pi n}}.$$

The series $\sum_n \frac{1}{\sqrt{\pi n}}$ is a Riemann series with exponent $\frac{1}{2} \leq 1$, hence *divergent*.

The walk is then *recurrent*: it returns almost surely to the origin.

- **Asymmetric case** ($p \neq \frac{1}{2}$). Then $4pq < 1$, and the general term $\frac{(4pq)^n}{\sqrt{\pi n}}$ is dominated by the geometric sequence $(4pq)^n$ with ratio < 1 . The series $\sum_n \frac{(4pq)^n}{\sqrt{\pi n}}$ is therefore *convergent*. The walk is then *transient*: with strictly positive probability, it never returns to the origin.

4.3 First passage at -1

For $r \in \mathbb{Z}$, write $\tau_r = \inf\{n \geq 1 : S_n = r\}$ for the first time at which the walk reaches level r . Here we study the law of τ_{-1} , that is, the probabilities $\mathbb{P}(\tau_{-1} = n)$ that the walk hits -1 for the first time at time n . Let us introduce the associated generating function:

$$F_1(s) = \sum_{n \geq 1} \mathbb{P}(\tau_{-1} = n) s^n, \quad s \in [0, 1].$$

Deriving the functional equation. Let us condition on the first step ξ_1 of the walk.

- If $\xi_1 = -1$ (probability q), the walk reaches -1 at the very first step: $\tau_{-1} = 1$.

- If $\xi_1 = +1$ (probability p), the walk is at $+1$ after one step. To reach -1 , it must then come down two levels, from $+1$ to -1 . Write τ_{-2} for the time needed to come down two notches starting from $+1$.

We thus obtain, for $n \geq 1$,

$$\mathbb{P}(\tau_{-1} = n + 1) = p \mathbb{P}(\tau_{-2} = n),$$

which translates on the generating functions into

$$F_1(s) = q s + p F_2(s), \quad \text{where } F_2(s) = \sum_{n \geq 1} \mathbb{P}(\tau_{-2} = n) s^n.$$

Now, by the spatial homogeneity of the walk, coming down from $+1$ to -1 amounts to chaining two successive one-notch descents, from $+1$ to 0 and then from 0 to -1 . These two durations are *independent* and each follows the same law as τ_{-1} ; in other words $\tau_{-2} = X + Y$ with X, Y independent and distributed as τ_{-1} . Since the generating function of a sum of independent variables is the product of the generating functions,

$$F_2(s) = F_1(s)^2.$$

Substituting into the previous relation, we arrive at the functional equation

$$\boxed{F_1(s) = q s + p s F_1(s)^2}.$$

Solving it. Setting $y = F_1(s)$, the equation becomes a quadratic equation in y (for $s \neq 0$):

$$p s y^2 - y + q s = 0, \quad \delta = 1 - 4 p q s^2.$$

For $s \in [0, 1[$, we have $4pq s^2 \leq s^2 < 1$, so $\delta > 0$ and there are two roots:

$$y = \frac{1 + \sqrt{1 - 4pq s^2}}{2ps} \quad \text{or} \quad y = \frac{1 - \sqrt{1 - 4pq s^2}}{2ps}.$$

The function F_1 is a power series vanishing at 0 (since $\mathbb{P}(\tau_{-1} = 0) = 0$). As $s \rightarrow 0^+$, the first root tends to $+\infty$ (numerator tending to 2, denominator to 0), which is ruled out. We therefore keep

$$F_1(s) = \frac{1 - \sqrt{1 - 4pq s^2}}{2ps}.$$

Power series expansion. We start from the generating function:

$$F_1(s) = \frac{1 - \sqrt{1 - 4pq s^2}}{2ps}.$$

We expand $\sqrt{1 - x}$ as a power series for $x \in]-1, 1[$:

$$(1 + x)^\alpha = 1 + \frac{\alpha}{1!}x + \frac{\alpha(\alpha - 1)}{2!}x^2 + \cdots + \frac{\alpha(\alpha - 1) \cdots (\alpha - n + 1)}{n!}x^n + \cdots$$

We apply it with $\alpha = \frac{1}{2}$ and replace x by $-x$, so that $(1 + (-x))^{1/2} = \sqrt{1 - x}$. The general coefficient of index $n \geq 1$ is

$$\frac{\frac{1}{2}(\frac{1}{2} - 1) \cdots (\frac{1}{2} - n + 1)}{n!} (-1)^n.$$

Turning the product into a binomial coefficient. The numerator is a product of n factors $\frac{1}{2} - j$ for $j = 0, \dots, n - 1$. Factoring $\frac{1}{2}$ out of each one, we write

$$\prod_{j=0}^{n-1} \left(\frac{1}{2} - j \right) = \frac{1}{2^n} \prod_{j=0}^{n-1} (1 - 2j) = \frac{1}{2^n} (1)(-1)(-3) \cdots (-(2n - 3)).$$

There are $n - 1$ negative factors (from -1 to $-(2n - 3)$), whence a sign $(-1)^{n-1}$, and the product of the absolute values is that of the odd integers $1 \cdot 3 \cdots (2n - 3)$. Multiplying and dividing by the even integers $2 \cdot 4 \cdots (2n - 2) = 2^{n-1}(n - 1)!$ so as to complete the factorial, this product of odd numbers equals $\frac{(2n - 2)!}{2^{n-1}(n - 1)!}$. We thus obtain an expression involving only *integer* factorials:

$$\frac{\frac{1}{2}(\frac{1}{2} - 1) \cdots (\frac{1}{2} - n + 1)}{n!} (-1)^n = -\frac{(2n - 2)!}{2^{2n-1} n! (n - 1)!}.$$

The coefficient of x^n in $\sqrt{1 - x}$ is therefore, for $n \geq 1$,

$$\sqrt{1 - x} = 1 - \sum_{n \geq 1} \frac{(2n - 2)!}{2^{2n-1} n! (n - 1)!} x^n.$$

We check the first few coefficients: $-\frac{1}{2}$, $-\frac{1}{8}$, $-\frac{1}{16}$, $-\frac{5}{128}, \dots$ Recognising $\frac{(2n - 2)!}{n! (n - 1)!} = \frac{1}{n} \binom{2n - 2}{n - 1}$, this can be rewritten with a binomial coefficient:

$$1 - \sqrt{1 - x} = \sum_{n \geq 1} \frac{1}{n} \binom{2n - 2}{n - 1} \frac{x^n}{2^{2n-1}} = \sum_{n \geq 1} \frac{2}{n} \binom{2n - 2}{n - 1} \frac{x^n}{4^n}.$$

Application to $x = 4pq s^2$. The substitution makes the factor 4^n disappear:

$$\frac{(4pq s^2)^n}{4^n} = (pq)^n s^{2n},$$

whence

$$1 - \sqrt{1 - 4pq s^2} = \sum_{n \geq 1} \frac{2}{n} \binom{2n - 2}{n - 1} (pq)^n s^{2n}.$$

Division by $2ps$ and identification.

$$F_1(s) = \frac{1}{2ps} \sum_{n \geq 1} \frac{2}{n} \binom{2n - 2}{n - 1} (pq)^n s^{2n} = \sum_{n \geq 1} \frac{1}{n} \binom{2n - 2}{n - 1} p^{n-1} q^n s^{2n-1},$$

and, after the reindexing $n \leftarrow n + 1$ in order to bring out the exponent $2n + 1$:

$$F_1(s) = \sum_{n \geq 0} \frac{1}{n+1} \binom{2n}{n} p^n q^{n+1} s^{2n+1}.$$

By identification of the coefficients,

$$P(\tau_{-1} = n + 1) = \frac{1}{n+1} \binom{2n}{n} p^n q^{n+1} = C_n p^n q^{n+1}$$

where $C_n = \frac{1}{n+1} \binom{2n}{n}$ is the n -th Catalan number, and $P(\tau_{-1} = 2n) = 0$ by parity, this last equality expressing the parity constraint: reaching the odd level -1 requires an odd number of steps.

Chapter 5

A problem of Euler

5.1 Introduction

Let us split the interior of a convex pentagon into three triangles by drawing two diagonals that do not meet. This can be done in five different ways. In a convex hexagon, one can draw 14 different sets of non-crossing diagonals splitting the interior into triangles. The problem we are going to tackle is a hard problem posed in the eighteenth century by the great Swiss mathematician Euler:

In how many different ways can a convex polygon with n sides be divided into triangles by drawing diagonals that do not meet?

Write c_n for the number of ways of splitting a polygon with $(n+2)$ sides into triangles. Thus, for pentagons $c_3 = 5$, for hexagons $c_4 = 14$, and so on.

5.2 Deriving the recurrence relation

Let us set $c_0 = 1$, which makes sense since there is only one way of decomposing a triangle, as there is nothing to do: there are no diagonals (a flat triangle).

Consider a convex polygon with $(n + 3)$ sides, which gives rise to c_{n+1} decompositions. Denote this polygon by $A_1A_2A_3\dots A_{n+2}A_{n+3}$. In each of the decompositions, the side A_1A_{n+3} of the polygon appears as one of the sides of some triangle of the decomposition. The third vertex of such a triangle is one of the vertices $A_2, A_3, \dots, A_{n+2}$. Let us look for the number of decompositions in which this third vertex is A_{k+2} . Letting k take successively the values $0, 1, 2, \dots, n$, we will obtain c_{n+1} by adding up all the numbers of decompositions thus obtained. The diagonals drawn from A_1 and A_{n+3} to the vertex A_{k+2} form a triangle which, inside the $(n + 3)$ -gon, delimits a polygon $A_1A_2\dots A_{k+2}$ with $k + 2$ sides and a polygon $A_{k+2}A_{k+3}\dots A_{n+3}$ with $(n - k + 2)$ sides. Consequently, these polygons can be decomposed in c_k and c_{n-k} ways respectively. We obtain a decomposition of the $(n + 3)$ -gon by combining each of the c_k ways of decomposing one of the polygons with each of the c_{n-k} ways of decomposing the other. Adding up over all possible values of k , we obtain

$$c_{n+1} = \sum_{k=0}^n c_k c_{n-k}. \quad (I)$$

We thus obtain a recurrence relation, on which we shall use generating functions (which are power series) in order to find the value of each c_i for every $i \in \mathbb{N}$.

5.3 Solution using generating functions

Let C be the power series in x whose coefficients are the unknown numbers c_i :

$$C(x) = \sum_{n=0}^{\infty} c_n x^n.$$

Squaring C , we obtain, by means of the Cauchy product,

$$C^2(x) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n c_k c_{n-k} \right) x^n \stackrel{\text{by (I)}}{=} \sum_{n=0}^{\infty} c_{n+1} x^n$$

Multiplying by x :

$$xC^2(x) = \sum_{n=0}^{\infty} c_{n+1} x^{n+1} \stackrel{\text{after reindexing}}{=} \sum_{n=1}^{\infty} c_n x^n = C(x) - 1$$

We thus obtain a functional equation to solve; setting $y = C(x)$:

$$xy^2 - y + 1 = 0$$

$$\Delta = 1 - 4x$$

We obtain two roots if $x \neq 1/4$:

$$y_1 = \frac{1 - \sqrt{1 - 4x}}{2x}$$

$$y_2 = \frac{1 + \sqrt{1 - 4x}}{2x}.$$

Letting $x \rightarrow 0^+$, we get $y_1 \rightarrow 1$ and $y_2 \rightarrow +\infty$, so we take the solution $y_1 = C(x)$.

Expanding $(1 - 4x)^{1/2}$ as a power series, we have

$$(1 - 4x)^{1/2} = \sum_{n=0}^{\infty} \frac{\frac{1}{2}(\frac{1}{2} - 1) \dots (\frac{1}{2} - n + 1)}{n!} (-4x)^n.$$

Two power series are equal if and only if they have the same coefficients, so:

$$c_n = \frac{\frac{1}{2}(\frac{1}{2} - 1) \dots (\frac{1}{2} - n + 1)}{n!} (-4)^n$$

Proceeding as in Chapter IV, Section 4.3, with the transformation of the product into a binomial coefficient 4.3, we then obtain

$$c_n = \frac{(2n)!}{n!(n+1)!} = \frac{(2n)!}{n!n!} \frac{1}{(n+1)} = \frac{1}{n+1} \binom{2n}{n}.$$

We thus recover the Catalan numbers, already met in Chapter 4 in the study of the random walk, even though the two problems have no apparent connection. This is no coincidence: the functional equation $C(x) = 1 + xC(x)^2$ obtained here is of exactly the same form as the one satisfied by the generating function $F_1(s) = qs + ps F_1(s)^2$ of the first passage times. The Catalan numbers appear precisely when an object decomposes into two independent sub-objects of the same nature, a structure which translates analytically into a quadratic term in the generating equation. It is this hidden kinship, invisible in the statements themselves, that power series make manifest.

Chapter 6

Conclusion

This internship, devoted to power series and their applications, first of all gave me an **analytic rigour** that I had not expected at the outset. Studying the behaviour of a power series at the boundary of its disc of convergence forced me to handle the notions of pointwise, normal and uniform convergence with care, and to write long and technical proofs, such as the one for Lowther’s series.

Beyond pure analysis, this work made me discover the **cross-cutting reach** of one and the same mathematical object. The generating function, which is nothing but a power series encoding a probability distribution, turned out to be a remarkably effective tool: it translates independence into a product, moments into derivatives at the point $t = 1$, and random composition into composition of functions. It is what allowed me to approach the Galton–Watson process and the random walk with the same weapons as the analysis of Chapter 1.

The fact that struck me most is undoubtedly the way in which **a problem with no apparent connection to analysis** reduces to a functional equation identical to that of a first passage time. Both lead to the same Catalan numbers. Seeing such a

striking coincidence emerge from a simple power series computation was, for me, the best illustration of the deep unity of mathematics.

On the practical side, this internship allowed me to **learn L^AT_EX**, whose usefulness goes far beyond this report (reports, CVs, cover letters). It also taught me a **critical use of artificial intelligence assistants**: these tools saved me valuable time in identifying sources – the papers of Fejér and Hardy–Littlewood, and Lowther’s work – but only on condition that their claims be checked systematically, since they are never reliable without cross-checking.

Finally, this work opens onto directions that I could only touch upon, in particular **applications in economics**: the discounting of an income stream, where the price of an asset is written as the sum of a series $P = \sum_n a_n$ whose value is computed using the same power series techniques. I have not yet found an application as striking as the Catalan numbers, but this line of enquiry has made me want to pursue the study of power series on the side of financial mathematics.

At the end of this internship, I am left with the feeling of having travelled a coherent path, from the most abstract analysis all the way to concrete applications, keeping a constant guiding thread: what happens at the boundary of the disc of convergence, at the point $t = 1$, is almost always where the essential information is hidden.